What access unlocks - and what it does not

Three rigorous economics papers · two applied development experiments and one classic disaster-economics read. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

This issue asks when **access is genuinely productive**. In rural India, access to either better classroom instruction or community study groups is insufficient by itself; the two inputs become effective when combined. In Ethiopia, removing application frictions opens factory employment and raises welfare briefly, but does not create a durable earnings ladder. Kenny's classic shows the same discipline at a systems level: safer construction matters, yet scarce budgets and weak enforcement make targeting and institutions as important as engineering. The broader trend is away from counting services offered and toward identifying complements, opportunity costs, and persistence.

Supporting Learning In and Out of School: Experimental Evidence from India

A cross-cutting cluster-randomized experiment across 200 villages in Assam asks whether an in-school Teaching at the Right Level intervention and parent-managed out-of-school study groups work better together than separately. Neither component alone raises learning significantly, but the combined program increases math scores by 0.11 standard deviations and language by 0.09, at modest cost. The complementarity is the interesting result; the principal caveat is that mechanisms and external validity remain less certain than the internally valid treatment contrast.

Do Factory Jobs Improve Welfare? Experimental Evidence from Ethiopia

The authors randomize light-touch application support for 687 young women seeking work in an Ethiopian industrial park, then follow them after eight months and four years. Support raises short-run employment, earnings, savings, and factory experience, but the earnings advantage disappears as the control group finds alternatives; early health costs also fade. Attrition is addressed with weighting and the paper reports multiple-testing adjustments and robustness checks. The result is a useful warning that entry frictions can be real even when the jobs unlocked are not durable ladders out of poverty.

Why Do People Die in Earthquakes? The Costs, Benefits and Institutions of Disaster Risk Reduction in Developing Countries

Kenny reframes earthquake mortality as an economic and institutional problem: engineering can prevent collapse, yet universal retrofitting may compare poorly with other life-saving investments in poor countries. Back-of-the-envelope benefit-cost calculations and cross-country evidence point toward targeted retrofits, simple enforceable codes, and oversight of public construction. The paper is a classic policy synthesis rather than a modern causal design, and several parameter values are uncertain, but its opportunity-cost and political-economy framing remains unusually useful.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Supporting Learning In and Out of School: Experimental Evidence from India*

Martina Björkman Nyqvist† Andrea Guariso‡

June 30, 2022

Abstract

This paper studies an education program targeting primary school children in rural India, which combines a standard in-school pedagogical intervention with an out-of-school study group program managed by parents. We rely on a cross-cutting experimental design across 200 villages and find the full program to significantly increase children's test scores in mathematics and language by 0.09 and 0.11 standard deviations respectively. When the two program components are implemented in isolation, there is no impact on children's learning. The cost-effectiveness analysis highlights high returns from adopting a multidimensional approach that supports children's learning processes both in and out of school.

JEL classification: C93, I21, O1

*We are grateful for comments and suggestions by Abhijeet Singh, Andy de Barros, Timothée Demont, Paul Glewwe, Selim Gulesci, Gregory F. Veramendi, and seminar participants at University of Munich (LMU), UC Merced, University of Goettingen, Aix-Marseille School of Economics (AMSE), University of Bologna, University of Milan-Bicocca, Trinity College Dublin, MWIEDC, EOS Development Webinar, YoDEV Webinar, as well as Fadhil Nadhif Muharam, Artur Obminski, Merve Demirel, and Harry Humes for invaluable research assistance. We also thank J-PAL South Asia and its staff, specifically Srijana Chandrashekhar; Mrignaina Tikku; Dwithiya Raghavan; Tanima; Akanksha Saletore; Maaike Bijker; Rithika Nair and Shobini Mukerji for support and management of the data collection and fieldwork. Finally, we thank Rukmini Banerji and Saveri Kulshreshth at Pratham India and Ingrid Eelde Koivisto at Pratham Sweden for insightful discussions about the programs and for their collaboration throughout the study, as well as for sharing administrative data. All mistakes are our own. Financial support from Carl Bennet AB, the Swedish Research Council (2016-05615) and the Mistra Foundation is greatly appreciated. Andrea also acknowledges funding from the TCD Arts and Social Sciences Benefactions (ASSB) Fund. The study was registered with the AEA RCT registry (0002817) and received ethical approval from IFMR Human Subjects Committee (IRB00007107) and TCD Ethic Review Board (05062018).

†Stockholm School of Economics and Misum.

‡University of Milan-Bicocca.

Introduction

The dramatic gains in school enrollments that have been achieved in low- and middle-income countries over the past two decades have not been mirrored by improvements in learning levels (World Bank, 2018; Altinok et al., 2018; Le Nestour et al., 2021). India is a case in point: more than 96% of all children in the age group of 6-14 years are enrolled in school, but the share of children in grade 5 that can read a grade 2 text or solve a grade 2 subtraction problem has stagnated below 50% for more than a decade (ASER, 2018). The Covid-19 pandemic and associated school closures further slowed learnings (Moscoviz and Evans, 2022; Björkman Nyqvist and Guariso, 2022), making the quest for effective approaches to boost students' achievements in low-income settings an even more pressing priority. Most programs studied so far in the literature focus on how to make schools more effective and the time spent in school more productive (see Muralidharan (2017) for a recent review). The role of further learning *outside* of the classroom has instead received far less attention.

This paper presents experimental evidence of a primary education program aimed at improving children's learning by combining a community-managed out-of-school program with a standard in-school pedagogical program. The out-of-school component is a *Study Groups* program, managed by community volunteers, which brings together primary school children of different ages to study and learn in groups outside of school hours. The in-school pedagogical program consists of *Learning Camps*, which are structured around the "Teaching at the Right Level" approach: students are temporarily re-arranged in the school based on their actual knowledge rather than on their grade, and teaching is tailored to their level.

We study the program's impact on children's learning using a randomized controlled trial with factorial design across 200 villages in Assam, northeast India. The villages were randomly assigned to one of the four study arms of equal size: full program (with both Study Groups and Learning Camps), Study Groups only, Learning Camps only, or control group. We conducted two main rounds of data collection: prior to the start of the program (mid-2018) and roughly 16 months after the program started (end of 2019). We surveyed the primary public schools located in each village (N=200), a representative sample of children enrolled within each school (N=5,726) and a representative sample of their caregivers (N=4,592).

We find that the full education program improved children's learning in math and language on average by 0.09 and 0.11 standard deviations (SD), compared to children in the control group. These effects translate into an increase in the share of children that achieve minimum standards (i.e. grade-2 knowledge) in mathematics and language, by 20% and 13% respectively. We also investigate the impact of direct exposure (attendance) to the program, using an instrumental variable approach (under the assumption that the program had no other effect on learning than through direct participation), and estimate learning gains between 0.38 and 0.48 SD.

We find no impact on children's learning levels when the two program components were implemented in isolation. The lack of impact of the in-school component (Learning Camps) is in contrast with previous assessments of the same pedagogical approach in Uttar Pradesh, where it resulted in large improvements in childrenâs learning (Banerjee et. al., 2016). We combine our data with data from the Uttar Pradesh study and can rule out a number of potential explanations for this difference in impact. However, we cannot directly identify the key driver(s), which highlights the challenges of translating successful programs into new contexts and over time.

Cost-effectiveness analysis shows that the full program, with both the in-school and out-of-school component, costs between 15 and 18.3 US$ per student per 0.1 SD increase in learning, which puts it in the middle of the cost-effectiveness distribution of 27 education programs assessed by J-PAL (Bhula et al., 2013). The analysis highlights that in the context of the ongoing expansion of the Teaching at the Right Level model across India and Africa, adding the out-of-school component can lead to significant gains at a very low additional cost.

Further analysis shows that timing of the in-school pedagogical intervention matters: by exploiting random variation in when the schools hosted the Learning Camps, we find that children exposed to camps earlier display higher test scores by the end of the study, suggesting that the camps put them on a steeper learning trajectory. We also find that all interventions increase parental engagement through more frequent interactions with the school, while the overall amount of time spent with children remains unchanged. At the school level, we find instead evidence of a substitution effect: schools that hosted the learning camps had a drop in educational inputs (both in terms of investments and interactions with stakeholders).

The existing body of evidence has mostly focused on in-school programs and highlights targeted pedagogical interventions as one of the most effective approaches (Glewwe et al., 2009; Banerjee et al., 2016; Glewwe and Muralidharan, 2016; Muralidharan, 2017). In our context, we find that the pedagogical intervention is only effective when paired with the out-of-school community program. Overall, the literature has paid little attention to continued learning outside of schools, despite the potential complementarities with what happens inside the school. There are few studies that look at the impact of after-school remedial education (Lakshminarayana et al., 2013; Banerjee et al., 2016; Chiplunkar et al., 2021; Romero et al. 2021), in some cases with the support of computer-assisted learning (Linden, 2008; Lai et al., 2015; Muralidharan et al., 2019). While results appear generally encouraging, most of these programs are implemented in schools right after (or before) official school

1.1. Intervention

The program studied in this paper target primary school children and includes two components, the Study Groups and the Learning Camps, implemented by Pratham Education Foundation.

Study Groups The community-based *Study Groups* provide a framework for the community to support out-of-school studying and learning, while at the same time maintaining costs and time commitment low, by harnessing the power of peer support. Pratham introduced the Study Groups in the target villages at the beginning of the school term and mobilized community volunteers, who then locally managed it. Each study group of about six primary school-age children was coordinated by one volunteer (typically a mother) and the total number of groups in the village varied depending on the number of children and volunteers. The structure of the study groups were flexible and it was up to each volunteer to set the frequency and length of the meetings. Pratham only conducted short monthly visits to the villages to distribute self-explanatory learning material that could be used to guide activities in the groups, while the groups were also encouraged to focus on other reading and studying activities, such as completion of school homework.[3]

Learning Camps The *Learning Camps* were implemented by Pratham in the schools during regular school hours and consisted of intensive bursts of teaching and learning activities focused on the foundational skills for reading and arithmetic. The camps were based on Pratham's pioneered pedagogical approach called Teaching at

[3]Study groups did not maintain any formal records of their activities but at endline we conducted a short qualitative survey with 40 volunteers across the study villages. There was significant heterogeneity, but most groups met 2 or 3 times per week for about 2 hours, and children mostly focused on the material shared by Pratham.

the Right Level (TaRL), which provides teaching that is tailored to a childâs individual learning level. Pratham introduced the learning camps in the target villages at the beginning of the school term and mobilized community volunteers to help with the implementation. The camps were then administered by Pratham staff with the support of available teachers and volunteers. The camps consisted of three periods of 10 days each (for a total of 30 days), spread over one teaching term (5 months). At the beginning of the camp, Pratham tested all children in grades 1 through 5, to identify their level of reading and arithmetic. Children were thereafter grouped according to their level and learned language and mathematics through specific activities and materials tailored to each group.

Full program The full program combines both the Study Groups and the Learning Camps components.

1.2. Sample and experimental design

The study sample of 200 villages was randomly selected from a list of villages in Nagaon district that Pratham identified as eligible.[4] Each village hosts one primary public school.

After baseline data collection, the 200 villages were randomly divided into four groups of equal size: *Learning Camps & Study Groups*; *Learning Camps*; *Study Groups*; *Control*. A second randomization was conducted to determine the timing of the implementation of the Learning Camps, given that Pratham could not run them at the same time across all villages due to resource constraints. Out of the 100 villages assigned to receive the Learning Camps (either alone or in combination with the

[4]Pratham's standard preliminary assessment considers official enrollment in the local primary school, accessibility of the village, and a qualitative assessment of the potential for mobilizing the community. Figure A.1 shows the spatial distribution of the study villages within Nagaon district.

Study Groups), we randomly selected (stratifying by group) half of them to receive the program right after baseline data collection (*phase I*), and the other half to receive it roughly 5 months later (*phase II*). Since the endline survey took place at the same time across the whole sample, this introduced random variation in children's time since exposure to the Learning Camps. Figure 1 illustrates the experimental design.

1.3. Data

Data for the analysis is collected from three sources: school, children, and their primary caregivers. The school survey included a combination of direct questions to the head teacher and observational records of school infrastructure. The child survey was administered to a random sample of 8 students in each grade 1 to 4 at baseline (thus expected to be in grades 2-5 at endline), for a total of 32 students per school.[5] Student selection was based on official school enrollment registries and sampled students who were absent from school on the survey day were surveyed at home. The child survey consisted of two parts: a test and a short survey. The test component included two tests, each one with a mathematics and a language section. The first test (ASER test) mirrored the standard test conducted yearly by the ASER Center across India for children aged 5 to 16. The second test (Test A) was instead created by the research team based on questions used in other studies in India (Muralidharan et al. 2020) and was designed to provide an alternative to the standard ASER assessment. Both ASER and Test A were conducted orally and individually with each sampled child by trained enumerators (see Appendix B for details). By endline Test A proved to be too easy for the students in the sample and displayed little variation. Therefore, in our main analysis, we focus on the standard ASER test

[5] Because some schools were smaller than the target, we eventually surveyed on average about 7 children per grade or 29 children per school.

and report the results on Test A (as well as on a comprehensive measure including both tests) in the appendix.[6] The student survey mainly contained questions about the childâs studying and learning activities. Finally, at baseline we randomly selected 6 children per grade (i.e. 24 per school) from our sample and tracked their primary caregiver (i.e. the person who dedicates the most time to looking after the needs of the child) at home.[7] The survey collected basic information on household characteristics and education practices.

The baseline survey was conducted between May and August 2018. Implementation of the programs started immediately after baseline data collection. The implementation of Learning Camps was rolled out in two phases, while the community Study Groups activities started in all villages immediately after baseline and continued until the end of the study. A short compliance survey was conducted in May and June 2019. The endline survey was conducted between November 2019 and January 2020, 16 months after the start of the program.

1.4. Baseline balance and validation

The schools in our sample had on average 53 students enrolled, 2 classrooms, and 4 teaching staff at baseline. Attendance on survey day was on average 70%. About one in four students (25%). reported never studying after school. In terms of learn-

[6]ASER test has the advantage of being comparable to national tests as well as to previous studies in the literature. One concern is that Pratham's Learning Camps, by dividing students according to the results on an initial ASER-style test, might be especially good in teaching to the ASER test. However, the lack of impact in the study arm exposed to Learning Camps only alleviates this concern. As reported in Appendix, our results are confirmed when considering the Test A mathematics scores, while they are typically non-significant when considering the Test A language scores, where half of the students correctly answering 90% of the questions (the average was 76%). Finally, results are confirmed when we combine both our tests in a single measure, using a two parameter logistic (2PL) item response theory (IRT) model (Jacob and Rothstein, 2016).

[7]The respondent was typically the mother (85%), followed by the father (5%), and the grandmother (3%). Because some schools were smaller than the target, we surveyed on average 23 primary caregivers in each village (see Table A.2 in appendix)

ing levels at baseline, the average student in our sample scored between level 1 (beginner) and level 2 (corresponding to 1-digit number recognition in math and letter recognition in language) in the ASER test components. While there is a progression in learning across grades (Figure A.3 and A.4 in Appendix), the overall learning levels are low: about 35% of students in grade 4 were unable to solve subtractions and 66% were unable to read a story, which are the expected standards for a grade 2 student. Table A.1 in Appendix A shows that observable characteristics of schools, children, and primary caregivers were balanced across study arms at baseline.

At endline we were able to track back 93% of the baseline sample. Attrition rates did not differ across study arms and there was no difference across treatment arms in characteristics of children lost at endline (Table A.3 in Appendix A).[8]

Program implementation and study design compliance are checked in two ways. First, we matched our sample of schools and children with detailed Pratham's administrative records to confirm that Study Groups and Learning Camps took place in every study village. We recorded correct adherence to the original design, with the exception of two villages with similar names, originally assigned to the control and the full program intervention, which had been mistakenly swapped by Pratham. In the analysis, we adopt a conservative approach and rely on the original random assignment.[9] The records show that Study Groups villages had on average 5 study groups and that within Learning Camps villages 93% of the children in our sample attended at least one day of Learning Camp in the schools (with an average attendance of 20 days). Second, at the endline we asked head teachers, children, and caregivers about interactions with Pratham and exposure to the programs. Such

[8]At endline one head teacher from the Study Groups treatment arm refused to answer the endline questionnaire (while still giving permission to test and survey children in the school) and we therefore miss one school survey. Caregivers' attrition mirrors children's attrition (93%).

[9]Using the actual assignment leaves in any case our results virtually unchanged.

recall information is likely noisy, but we find the distribution of the replies across study arms to mirror the study design (Table A.5 and A.6 in Appendix).

Overall, our checks on baseline balance, attrition, and implementation alleviate possible concerns related to the integrity of the study design and hence, differences in outcomes at endline should be attributed to the program.

1.5. Empirical Strategy

Our main estimating equation is:

$$Y_{i,v} = \alpha_1 LC\&SG_v + \alpha_2 SG_v + \alpha_3 LC_v + \Omega X_{i,v} + \epsilon_{i,v} \tag{1}$$

where $Y_{i,v}$ is the outcome of child i, attending school in village v. $LC\&SG$, SG, and LC, are indicator variables that take value one if the village was assigned to the full program (i.e. both Study Groups and Learning Camps), to the Study Groups alone, or to the Learning Camps alone, respectively. $X_{i,v}$ is a vector of individual-level control variables: age, gender, indicators for grade, and baseline value of the outcome variable $Y_{i,v}$. Standard errors are clustered at the village level. School and household level analysis follows a similar specification, with variables defined at the corresponding level. Finally, to form a judgment about the impact of the intervention on a family of n related outcomes, and address potential multiple hypothesis testing concerns, we follow Kling et al. (2004) and estimate a seemingly unrelated regression system to derive the average standardized treatment effect (ASTE).

2. Results

2.1. Student Learning Outcomes

Table 1 presents our main results of the impact of the program on children's learning outcomes. We find that the full program significantly increased mathematics and language test scores by on average 0.11 and 0.09 SD, respectively, compared to the control group.[10] The average effect size of 0.1 SD corresponds to the median effect size found across 270 different educational programs implemented in low- and middle-income countries, recently reviewed by Evans and Yuan (2021). In our context, it translates to an increase of 20% (13%) of primary school children that achieve minimum standards (i.e. grade 2 level) in mathematics (language) (columns 2 and 4).

When studying the impact of the two program components (Learning Camps in the schools and Study Groups in the villages) implemented individually, we do not find any increase in children's test scores, irrespective of the subject. Not only are the results not statistically significant, but the estimated coefficients are close to zero. The p-values confirm that the full program led to a significantly higher impact on children's test scores than either of the two components alone, with the differences being significant at the 10% level.[11]

The lack of impact of the Learning Camps component alone is in contrast with the results in Banerjee et al. (2016), which look at the same pedagogical model in Uttar Pradesh. To investigate this further, we combine their data with ours.[12] The

[10]We follow the literature (e.g., Banerjee et al., 2016) and normalized ASER test scores using the mean and standard deviation for the control group.

[11]Figure A.5 in Appendix A shows the change in learning levels experienced across the different study arms, while Figure A.6 and A.7 report the endline distribution of the different learning levels by grade and study arm.

[12]We thank the authors for making their dataset available to us.

(2) $$Y_{i,v} = \beta_1 \widehat{LC_{att}\&SG_{att}}_{i,v} + \beta_2 \widehat{SG_{att}}_{i,v} + \beta_3 \widehat{LC_{att}}_{i,v} + \Omega X_{i,v} + \varepsilon_{i,v}$$

Where $\widehat{LC_{att}\&SG_{att}}$ takes value one if student i attended at least one day of the Learning Camp in the school and reported attending at least once the Study Groups in the village, $\widehat{SG_{att}}$ takes value one if the student reported attending the Study Groups, but never attended the Learning Camps, and $\widehat{LG_{att}}$ takes value one if the student attended the Learning Camps, but reported never attending the Study Groups. We instrument the three endogenous attendance variables with the random assignment to the different study arms.[16] The 2SLS estimates in Table 2 (columns 3 to 4) indicate that the program had a large impact on children who attended both the Learning Camps in the schools and the Study Groups in the villages, with learning gains ranging between 0.38 and 0.48 SD.

2.2. Cost Effectiveness

We use data from Pratham to calculate cost-effectiveness for each of the intervention arms, estimating the cost of serving 50 villages with the full program, Learning Camps only, and Study Groups only. There are a few indications emerging from the cost analysis (Table 2). First, as expected, the Learning Camps component is significantly more expensive (60% more) than the Study Groups component. Second, there are strong cost synergies that can be exploited by combining the two components. The average cost for the Learning Camps only program was 15.9 US$ per student, for the Study Groups only program was 10 US$ per student, and it was 17.2 US$ per

[16]We make the (strong) assumption that the program only affected learning through direct participation. The most likely direction of the potential bias from violating the exclusion restriction is towards underestimating the true effect of direct exposure, as children not directly exposed might benefit from interactions with exposed children.

student for the combined program. This means that in the context of an expansion of the Learning Camps program, which we are currently witnessing across India and the African continent, it costs very little to add the study group components, with potentially very large returns.

Using the learning gains reported in Table 2 and the two cost-effectiveness numbers, we find the cost of increasing learning outcomes by 0.1 SD to range between 15 US$ and 18.3 US$ per student for the full educational program which implies an impact between 0.55 SD and 0.67 SD per 100 US$ investment (under the simplifying assumption of linear returns). To put these numbers in perspective, the program falls right in the middle of the distribution of a set of 27 education programs aimed at improving learning outcomes and evaluated using randomized controlled trials for which J-PAL has assessed the cost-effectiveness (Figure A.8 in Appendix).

2.3. Heterogeneity

We randomly assigned schools to host the Learning Camps in different phases, with a 5-month difference in time. Columns 1 and 2 of Table 3 report the results when we estimate equation (1) and interact the indicators for assignment to a Learning Camps study arm (*LC&SG* or *LC*) with an indicator for being in the first implementation phase. The estimates show that children in schools that hosted the Learning Camps earlier achieved higher test scores by endline. Although imprecisely estimated, the results suggest that the full education program led to learning improvements twice as large for students who received the program earlier on.[17] This result is consistent with the dynamic complementarity hypothesis of skill formation (Cunha and Heckman, 2007): Learning Camps managed to put children on a steeper learning trajectory, whereby they could cumulate more knowledge over time and, hence,

[17] Figure A.9 in Appendix visually illustrates this difference.

dren's learning process in school, such a substitution effect might have undermined the effectiveness of the program.

3. Conclusion

This paper provides the first experimental evidence on the impact of an educational program that combines a community-managed out-of-school component with a standard in-school pedagogical intervention targeting primary school children.

We find that the program was successful in raising children's learning levels in both mathematics and language by 0.1 SD on average, over a period of 16 months. The result shows that the combination of the two components was important: when the programs were implemented alone, neither of them had any impact on children's learning. These findings speak to the multi-dimensionality of the learning process and show that educational programs that intervene on several dimensions at the same time can take advantage of complementarities (also in terms of costs) and lead to significant learning gains for the children. These findings also highlight the important role that out-of-school learning programs can play, despite having received little attention so far both in policy and in the literature. We estimate the full program to be quite cost-effective and the implementing organization, Pratham, recently made it its new flagship educational model across India. Our results also indicate that organizations should test ways to increase program participation, in order to ensure that the benefits of the programs are fully grasped by the target beneficiaries.

We also examine how program impact evolves over time, which is something quite rare in the education literature, where only 1 in 10 studies is assessed more than one month after the intervention ended (McEwan, 2015). We do so both by exploiting random variation in the timing of the implementation within our original sample, finding estimates that are significantly different from our average impact. This shows that short-term evaluations of education programs, which represent the norm in the literature, are likely to provide a very incomplete assessment of the programs on children's outcomes.

The lack of impact of the pedagogical, Leaning Camps program, when implemented in isolation, is in contrast with previous findings in the literature which further highlights the importance of reviewing and adapting even the most successful programs to the local contexts. A recent study by Duflo et al (2021) provides a step in this direction, by testing alternative models of the Teaching at the Right Level approach in Ghana, where they find impacts between 0.08 SD and 0.15 SD. Our results suggest an alternative way to enhance the effectiveness of the traditional Learning Camps model, by combining it with an out-of-school, community-led, study groups program.

Future research should investigate how the NGO and evaluation teams best can work with a program adaptation approach when scaling up programs (such as the Teaching at the Right Level program) in order to tweak and modify the program along the implementation phase to fit the new context. More in general, more research is needed to study out-of-school programs where parents and community are engaged in the children's learning process. Finding effective ways to improve uptake and engagement of the community and parents to increase local ownership of these initiatives and facilitate their scaling up is an important next step.

The World Bank Economic Review, 39(1), 2025, 124–142
https://doi.org10.1093/wber/lhae015
Article

Do Factory Jobs Improve Welfare? Experimental Evidence from Ethiopia

Girum Abebe, Niklas Buehren, and Markus Goldstein

Abstract

This study explores the impact of a light-touch job-facilitation intervention that supported young female job seekers during the application process for factory work in a newly constructed industrial park in Ethiopia. Using data from a panel of 687 job seekers and randomized access to the support intervention, the study finds that treated applicants are more likely to be employed and have higher earnings and savings eight months after baseline, although these impacts are short-lived. Four years later, the effects on employment and income largely dissipated. The results suggest that young women face significant barriers to engaging in factory work in the short run that a simple job-facilitation intervention can help overcome. In the long term, however, these jobs do not offer a better alternative than other income-generating opportunities.

JEL classification: J24, J28, J63, O14

Keywords: Ethiopia, industrial park, factory jobs, female employment, income

Public Disclosure Authorized

1. Introduction

A large body of empirical evidence suggests that stable wage work in formal firms is a preferable path to prosperity for most labor market entrants and that job seekers queue for wage work in developing countries (Blattman and Dercon 2018; Contreras, Roberto, and Esteban 2017; Meghir, Narita, and Robin 2015; Weiss 1980). These jobs are often considered better than most poor people's alternatives because they offer a higher degree of economic security (Roy 2004). Formal firms, especially in the industrial sector, often pay higher salaries than informal employers because of efficiency wages, exporter wage premia, firm competition for scarce skills, and union bargaining (Bernard and Jensen, 1999); El Badaoui, Strobel, and Walsh 2008; Verhoogen 2008). More recent evidence indicates that industrial jobs can lead to welfare improvements, especially for female workers (e.g., Getahun and Villanger 2018; Suzuki, Mano, and Abebe

Girum Abebe (corresponding author) is a senior economist at the World Bank, Washington, DC, USA; his email address is gtefera@worldbank.org. Niklas Buehren is a senior economist at the World Bank, Washington, DC, USA; his email address is nbuehren@worldbank.org. Markus Goldstein is a lead economist at the World Bank, Washington, DC, USA; his email address is mgoldstein@worldbank.org. The authors are grateful to the World Bank Umbrella Fund for Gender Equality for funding the research. They thank Taylor Van Salisbury, Adiam Hagos Hailemicheal and Endale Gebre Gebremedhine for excellent research assistance and the Ethiopian Development Research Institute (EDRI) for expert field implementation. This paper is an output of the World Bank Africa Gender Innovation Lab. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development / World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent. A supplementary online appendix for this article can be found at *The World Bank Economic Review* website.

2018). There is also evidence suggesting that industrial work can have knock-on effects on women's empowerment, demand for own and children's education, household bargaining power, and perceived quality of life (Heath and Mobarak 2015; Jensen 2012; Kabeer 2002).

Against this background, Ethiopian policy makers have prioritized large-scale industrial development centered on attracting foreign capital investments in export-oriented and labor-intensive industries that are concentrated in industrial parks to drive job creation (NPC 2015).[1] Jobs in industrial parks offer wage-employment opportunities especially for young women with relatively low educational attainment, limited job-search experience, and often no history of formal employment (Meyer et al. 2021). To study the welfare impacts of these jobs, this study randomized access to job-application support in partnership with three firms in one of the newly built industrial parks, Bole Lemi.[2] In doing so, it tests the hypothesis that, by supporting and facilitating the job application process for young women who seek entry-level production line positions, the intervention reduces job search barriers and improves labor market outcomes.

More specifically, the intervention is designed to overcome spatial and informational frictions during job search that can trap young women in unemployment or low paying self-employment activities locally, a concern that will increasingly become critical as cities expand. Following urbanization in Nigeria, Animashaun and Emediegwu (2023) show that aggregate household labor supply declined impacting women more adversely compared to men. In Ethiopia, Abebe et al. (2021) find that the individual's likelihood of formal employment is negatively related to their home's distance from the city center and that an intervention subsidizing transport costs can reverse the tendency to engage in unproductive self-employment in favor of formal wage employment. That study also found that young unemployed people spent close to 16 percent of their limited income on search-related activities in Addis Ababa. Deserranno (2019) shows that financial incentives increase the applicant pool for health-care positions in rural Uganda. However, local recruiters may not have a good understanding of the prohibitive role of application cost. Abebe, Caria, and Ortiz-Ospina (2021) show that firms significantly underestimate the positive impact of job-application incentives in improving both the quantity and quality of applicants. The findings show that subsidizing applications is particularly beneficial for young women who stand to gain the most from the employment opportunities.

Furthermore, young women may not necessarily search actively and in a targeted way—instead, they tend to primarily use informal search channels and are likely detached from the labor market. Job boards, where job advertisements for formal jobs are commonly posted, mostly cater to those with college-level education and rarely contain information suitable for less educated and experienced women and youth (Franklin 2018). As a result, the labor market this paper explores is highly segmented by gender and education levels, largely excluding uneducated youth and women from formal employment.[3]

Similarly, lack of information about available job opportunities also precludes effective job search. For example, Ashraf et al. (2020) find that making career benefits salient on recruitment posters increases the quality of applicants for a health-care position in Zambia. In the study sample, only 15 percent of job seekers had heard of the Bole Lemi industrial park, about 75 percent had no prior factory experience, and only 1.7 percent had any experience working in the industrial park at baseline. Lacking job-specific experience, job seekers may be unfamiliar with basic yet essential aspects of the job-search process including the job-application and interview process. As a consequence, the median applicant in this study made no

1 The construction of a new industrial park specializing in the textile, garment, and leather industries was primarily meant to achieve two highly publicized goals: job creation and generating earnings from exports.

2 All partnering firms were foreign owned and produced finished garments for export. They also had large-scale hiring plans for the study duration.

3 This is reflected by a very high unemployment rate reaching respectively 34.1 percent and 23.7 percent among primary and secondary education completed, while the unemployment rate among post-secondary education is 16.3 percent in Ethiopia (CSA 2016). In particular, the unemployment rate is the highest among young women with secondary education reaching as high as 36 percent in urban areas (CSA 2018).

job applications for a formal wage-paying job in the four weeks leading up to the baseline survey. Current understanding of which policies can effectively help young women secure formal jobs remains limited. Yet tackling the barriers that prevent young women from participating in the formal labor market is a first-order problem for many low-income countries (Dinkelman and Ngai, 2022).

This study's research examines the impacts of a job-facilitation intervention eight months and four years after the initial job screening and application phase. The study finds that the extra support during the application process increases individuals' likelihood of being employed eight months after the treatment. Higher employment levels in wage work, in particular factory work, appear to be the driving force behind this result. This finding suggests that young women face significant barriers to labor-market participation and that a simple, one-off facilitation, support improves their transition to formal employment in the short run. The intervention also raises reported monthly income by nearly 30 percent. Higher income in the treatment group is accompanied by a 16 percent increase in nonfood expenditures and a 57 percent increase in savings. However, the study also finds an adverse impact on health outcomes and a downward adjustment of job applicants' expectations and perceptions regarding the quality of factory work in response to the treatment.

Many of these effects were short-lived. In the long run—that is, four years after the initial job screening—those who had received job-application support were more likely to have had factory work experience and factory employment experience, but there were no longer any differences in earnings, expenditures, and savings; mainly because the control group caught up with the treatment group in terms of finding alternative employment opportunities. For example, average weekly number of hours worked eventually converged—67 hours for treatment and 66 hours for control individuals. Likewise, the difference in average monthly earnings between women in the treatment (1,474 birr) and control (1,350 birr) groups is no longer significant.[4]

The results from the four-year follow-up survey also show that the adverse health impacts for women in the treatment group disappear over time and that factory work is considered no more physically demanding or unhealthy than other types of jobs individuals engage in. The downward adjustment of perceptions and expectations regarding the quality of factory work, however, is maintained in the long run. This is consistent with the fact that only 12.2 percent of young women who were assigned to the treatment group and started factory jobs were still working in factories four years later. These findings suggest that factory jobs may fall short of providing a wage premium and offer only limited potential for upward mobility in the long run.

This paper makes three important contributions to the literature. The first contribution is that, to the best of the authors' knowledge, this is the first study to systematically evaluate job-facilitation support in the context of industrial parks in low-income countries. Industrial parks host firms in modern production facilities that are often located near each other but far from urban centers and residential areas. These features of industrial parks have several implications for job searches, employment, and worker welfare. First, the relative isolation of industrial parks from city centers creates a disconnect between job opportunities and job seekers. Inadequate access to information and to transportation infrastructure exacerbates the effects of this spatial disconnect. As described earlier, only one in eight job seekers in the study sample had ever heard of the Bole Lemi industrial park, and only 1.7 percent had any experience working in the industrial park at baseline. In the presence of such informational and spatial frictions, job seekers' beliefs about their own labor-market prospects and job attributes may be inaccurate (Banerjee and Sequeira 2020).[5] By providing job seekers with information and transportation to industrial parks, the intervention addresses these barriers to an effective job search. Second, industrial park jobs are in export-oriented

4 At the time of the baseline survey in July 2016, US$1 was worth nearly 22 birr.

5 The recruitment and worker-retention challenges that the newly established Bole Lemi and Hawassa industrial parks face highlight these constraints (Hardy and Hauge 2019).

sectors, commonly in garments and textiles, that have seen growing scrutiny and intense buyer audits to ensure fair employment practices and workplace compliance. Yet it remains unclear whether industrial park jobs are of better quality than alternative employment options. Third, factories in industrial parks predominantly employ young migrants in low-skill, low-wage positions with limited unionization and access to local safety nets (Meyer et al. 2021).

The second contribution of the study is linked to the literature on active labor-market policies, especially work that explore search frictions in low-income countries (Abebe, Caria, and Ortiz-Ospina 2021; Abel et al. 2019; Franklin 2018). In growing cities, spatial and informational frictions are critical for the allocation of limited job opportunities. As a result, job search is often a self-managed process constrained by the availability of information and affordable transportation options—to acquire information about job opportunities, job seekers commonly commute to city centers to visit employers, work sites, or general-purpose job boards where vacancy information is physically displayed (Franklin 2018). Typically, applications are also submitted in person and involve commuting to submit CVs and go to interviews, written exams, and medical tests. Young women often face challenges related to accessing affordable and safe transportation options to make such commutes. This paper tests whether light-touch job-application support can help ease search frictions and improve the labor-market prospects of inexperienced young women. The study finds that job-search frictions deter female job seekers' entry into industrial jobs, but that the search frictions do not generate long-term negative impacts on their labor-market outcomes. This result is largely consistent with emerging evidence showing that lowering search cost increases job-search intensity and improves employability in the short run, but has no long-lasting impacts (Abebe et al. 2021).

The third contribution is that this study provides a direct test of whether jobs in industrial parks provide young women with a reliable pathway to economic empowerment in a setting in which severe gender gaps in economic outcomes persist. As more industrial parks are planned and constructed in the country, understanding, and documenting the impact of these new manufacturing jobs on the livelihoods and empowerment of young women—and the associated economic and social impacts on young women's households—is crucial for shaping more inclusive industrialization policies.

This study is most closely related to the work of Blattman and Dercon (2018) and Blattman, Dercon, and Franklin (2022), who compare the one- and five-year impacts of offering young job seekers in Ethiopia factory work or a cash grant versus a control group. This study differs from theirs in that they offer all job seekers in their wage work treatment arm a factory job. This study randomizes light-touch job-application support, which, we believe, makes the study more applicable to the actual choice set of policy makers and offers a greater scalability potential. In essence, this study examines search frictions that prevent job seekers from obtaining industrial jobs. The focus of Blattman and Dercon (2018) and Blattman, Dercon, and Franklin' (2022) is on studying matching frictions and overcoming liquidity constraints. In addition, the applicant population in this study is composed entirely of women with about 70 percent migrants, whereas their sample contains both men and women and mostly nonmigrants.[6] Finally, this study is situated within the context of industrial-park factory jobs offered by foreign-owned exporting firms in the garment and textile sector in an urban environment, whereas they sample workers from four domestic and one foreign firm in five sectors in four administrative regions. Despite these differences, the key finding that factory jobs are not economically more beneficial than alternative employment opportunities in the long run is consistent with their results.

The remainder of the paper is organized as follows. Section 2 discusses the context of factory work in Ethiopia. Section 3 describes the research design, data, and estimation strategy. Section 4 presents and discusses the impact estimates of the intervention on a range of outcomes related to employment, earnings, and health. Section 5 provides concluding remarks.

6 The migration status of the study sample at baseline is not discussed in Blattman, Dercon, and Franklin (2022).

2. Industrial Parks and Factory Jobs in Ethiopia

Despite accelerated progress toward economic development and poverty reduction, Ethiopia remains among the poorest countries in the world. Its rapidly growing population requires large-scale job creation to meet the demand for economic opportunities. Informal work (own account and family employment) dominates employment, with only 13.5 percent of the workforce classified as wage workers (ILO 2018). With over 70 percent of the population engaged in the agricultural sector, structural transformation has been slow (Bezawagaw et al. 2018). The country's growing population of landless youth in rural areas motivates the government's plan to transition from reliance on agriculture to manufacturing, particularly of tradable goods. Growth in export-oriented industrial production is one way to absorb large portions of youth looking for employment (FDRE 2010; NPC 2015). The construction of industrial parks is the main policy tool that the Ethiopian government uses to achieve this goal.[7]

There is significant evidence from other contexts that this strategy may pay off. Formal employment, particularly in export-oriented sectors, can offer workers and in particular young women better working conditions, greater workplace safety, and higher wages (El Badaoui, Strobel, and Walsh 2008; Verhoogen 2008, Lopez-Acevedo and Robertson 2016; McCaig and Pavcnik 2018; Tanaka 2020). There are several ways in which the export sector supports job growth and helps to improve working conditions. First, exporting fuels worker reallocation from informal enterprises to more productive firms that potentially offer higher monetary and nonmonetary remuneration. McCaig and Pavcnik (2018), for example, find that a drastic decline in U.S. tariffs on Vietnamese exports in 2001 led to a significant increase in formal-firm employment, particularly for younger workers from exporting provinces. Fukase (2013) shows that the tariff decline was also associated with comparatively greater wage growth among unskilled workers in provinces with greater exposure to export. Similarly, following textiles and apparel exporting, Lopez-Acevedo and Robertson (2016) find that low-skilled women across several countries in South Asia received higher wages in industrial employment compared to agriculture.

Second, supporting export-oriented sectors by place-based policies, such as industrial parks, can attract foreign direct investment that improves employment and wages. Lu, Wang, and Zhu (2019) and Zheng et al. (2017), for example, show that industrial parks attracted a new wave of foreign investment that stimulated local production and generated significant spillover effects in China. Further, both studies independently show that firms in special economic zones generated more employment and paid higher wages compared to firms outside the zones. Similarly, trade policies that promote economic integration can facilitate the entry of greenfield investments. McCaig, Pavcnik and Wong (2022) look at the longer-term cumulative effect of U.S. tariff cuts on firm entry and changes in employment in Vietnam. The authors find that the tariff cuts had long lasting impacts on labor demand and employment particularly in industries that experienced the largest decline in U.S. tariffs.

Third, export-oriented firms operate in globally competitive supply chains and face increasingly more stringent labor standards and labor-compliance requirements. To meet these standards, export-oriented firms often improve working conditions and worker compensation. Tanaka (2020) finds that exporting had significant positive effect in improving the working conditions of garment workers in Myanmar even when there was no impact on wages.

The African experience with export-oriented industrial parks, however, has been mixed (Bräutigam and Xiaoyang 2011; Farole 2011). Even where industrial parks have successfully attracted investment, concerns remain over the quality of employment, labor rights, and issues of sustainability. Labor-intensive industries that are typically found in industrial parks often produce low-value commodities with thin profit margins. With narrow margins, the international competitiveness of these industries mainly lies in minimizing production costs along the value chain by adopting measures that potentially diminish the

7 As stated in Industrial Parks Proclamation No. 886/2015, industrial parks are established to "upgrade industries and generate employment opportunities."

quality of employment. Well-organized firms in industrial parks also possess market power that enables them to use collective action and jointly set wages and control labor mobility within the park, practices that are detrimental to workers' welfare.

In Ethiopia, the first industrial park, Bole Lemi, was completed in 2014.[8] Located on the outskirts of the capital, Addis Ababa, Bole Lemi hosts companies engaged in garment and leather goods production. These companies are all foreign-owned and produce almost exclusively for the export market. At the initiation of the present study, 10 firms were operating in Bole Lemi industrial park.[9] Most firms in the industrial park had just begun production and were hiring workers in great numbers to gradually increase capacity. These firms were primarily hiring female employees and thus creating a path for many of these young women to enter the formal labor market for the first time. In Ethiopia—where women are at an inherent disadvantage because of social norms, unequal institutions, and other barriers to labor-market participation—the light-manufacturing sector has the potential to offer them a path out of poverty and into economic independence. Emerging evidence suggests that industrial employment generates welfare improvements for female workers in the country's modern horticultural sector (Getahun and Villanger 2018; Suzuki et al. 2018). Moreover, evidence from other developing countries shows that low-skill manufacturing jobs can promote gender equity and improve quality of life for female workers, as well as their children's health and educational outcomes (Hewett and Aminb 2000; Nicita and Razzaz 2003).

Nevertheless, *feminization of labor* can have unintended consequences. For example, Ghosh (2009) and Seguino (2000a) show that East Asia's export-oriented industrialization was highly dependent on gender-based wage inequalities. Furthermore, segmentation of women into these export-intensive industries characterized by high price elasticity restricts their bargaining power. These forces allowed firms to keep wages artificially low and widened the gender wage gap (Berik, Rodgers, and Zveglich 2004; Chamarbagwala 2006; Pitt, Rosenzweig, and Hassan 2012; Seguino 2000b).

3. Research Design, Data, and Empirical Strategy

To uncover the short- and longer-run impacts of factory work on a series of welfare indicators, this study evaluates an intervention that used a relatively light-touch approach to support and facilitate the job-application process at firms located in the Bole Lemi industrial park for a randomly selected sample of eligible female job seekers.

3.1. Research Design

During the initial phase of the evaluation, the research team collected hiring plans from each participating firm and used this information to target interested job candidates. All partnering firms insisted on hiring exclusively women in a certain age range for the factory floor jobs considered for this study.[10] The factory positions were advertised through various channels, including posting advertisements in public places, passing out flyers in highly frequented areas of Addis Ababa, coordinating with youth associations, and using other forms of community mobilization. Unemployed individuals who had registered with their local *woreda* were also contacted directly.[11]

8 This industrial park is also known as Bole Lemi I. At the time of writing this paper, Ethiopia had 10 operational or semi-operational industrial parks (Adama, Bahir Dar, Bole Lemi I, Debre Birhan, Eastern, Hawassa, Jimma, Kilinto, Kombolcha, and Mekelle).

9 At the time of writing in February 2023, there were 10 factories. Eight of these factories produced textile and garments, and the remaining two processed leather goods.

10 These firms hired and employed men in other positions in areas such as logistics, building maintenance, security personnel, and managerial positions.

11 *Woredas* are the third-level administrative units after regions and zones **nationally**. In Addis Ababa, the city is divided into sub-cities and each sub-city is further subdivided into woredas.

During the recruitment process, potential candidates who were interested in factory employment were asked to bring identification and documentation of their qualifications to the nearest screening center, which were set up in several *woreda* offices across the three sub-cities of Addis Ababa. Trained enumerators staffed these screening centers during regular working hours to confirm that job applicants fulfilled the formal eligibility criteria for the advertised positions. Applicants with incomplete documentation or those who did not meet all the firms' eligibility criteria were screened out from the study. All applicants who met the eligibility criteria and could produce sufficient documentation were selected into the sample and asked to stay for the baseline survey.

Immediately after the baseline survey, study participants were randomized into treatment and control groups using a lottery, with two-thirds of applicants in the treatment group and one-third in the control group. Each treatment applicant was assigned to a firm for an interview. Since each participating industrial park firm had set its own hiring criteria, there were multiple distinct applicant groups.[12] Treatment applicants who were eligible for multiple firms were randomly assigned to one of the firms for which they were eligible.

Once applicants were assigned to a firm, the enumerators formally invited treatment individuals for the job interview and offered transportation to the Bole Lemi industrial park on the day of the interview.[13] Treatment and control individuals learned about their treatment status after completing the baseline survey but before they left the screening center. On the day of the interview, enumerators confirmed the treatment status before providing transport to the firm. After the interview process was completed, all candidates were offered transportation back to the screening center. The intervention did not prevent individuals in the control group from adopting alternative and commonly used pathways to applying for factory employment in any of the Bole Lemi industrial park firms.

Of those in the treatment group, 63.6 percent reported at the screening center on the day of the interview and were provided with transport to the industrial park. Firms conducted interviews with job candidates according to their own hiring procedures and decided whether to offer applicants the factory job. For some firms, this meant that candidates were required to complete additional stages in the hiring procedure such as a medical exam or an aptitude test. Nevertheless, the eligibility verification at the screening centers before baseline was effective as nearly all candidates were selected.[14] Successful applicants then had the opportunity to accept or decline the job offer. Of those who were interviewed and who were offered a position, 48.3 percent accepted the job offer.[15]

Applicants who accepted the job offer typically also had to go through an onboarding process at the firm, which mainly consisted of training that lasted for several days or weeks depending on the firm. This training focused mostly on technical skills related to fabric handling, stitching, pressing, packing, and quality control.

3.2. Data

Baseline data collection took place at screening centers directly after the eligibility checks but before treatment assignment to minimize attrition and ensure full comparability. All eligible job applicants who

12 For example, one firm accepts individuals aged 18 to 35, but another firm accepts only those aged 18 and 28, or with respect to education, one firm accepts individuals who have completed grade 5, but another accepts only those who have completed grade 8.

13 In the pilot of the evaluation design, one primary reason that fully eligible candidates did not attend their scheduled interview was that they had difficulty getting to and finding Bole Lemi I. Providing transportation to the interview was intended to minimize sample attrition at this stage of the hiring process.

14 In the days after the opening of Bole Lemi I industrial park, firms encountered significant challenges in finding suitable workers, and information on applicants' sex, age, and education was almost exclusively used to make hiring decisions.

15 After accounting for attrition, the corresponding figure in the balanced sample is 41 percent.

agreed to take part in the study were interviewed at baseline, and information from 935 respondents was collected between June and August 2016.

Approximately eight months and four years after the intervention, an effort was made to re-interview all baseline respondents face-to-face. A total of 827 baseline respondents were successfully tracked in the short-term follow-up survey that took place between January and March 2017—an attrition rate of nearly 12 percent. The long-term follow-up survey took place between February and April 2020, and 741 baseline respondents were interviewed—an attrition rate of about 21 percent. The 687 respondents interviewed in all three survey waves are at the core of the impact analysis. The attrition rate for this strictly balanced sample is nearly 27 percent.

Losing more than one quarter of respondents over a period of four years is of potential concern. Whether attrition systematically differs according to treatment status and whether it varies with any observable individual characteristics is tested in this paper. Table S1.1 in the supplementary online appendix, indicates that treatment is not correlated with attrition of respondents and that the sample composition did not meaningfully change between baseline and the two follow-up surveys because of attrition. In subsequent analysis, the study further accounts for attrition by weighting each observation by the inverse of its predicted probability of panel inclusion (being tracked at follow-up) using a leave-one-out logit regression in the first stage. It also applies several bounding approaches to examine the sensitivity of the findings to different assumptions regarding missing data caused by attrition.[16]

Table 1 indicates that study participants randomized into the control group were on average 25 years old at baseline, one year older than those in the treatment group. Approximately 30 percent of control respondents were married, and slightly more than one-quarter were mothers at baseline. The number of migrants among applicants for factory work was high; nearly 70 percent of respondents had migrated to Addis Ababa from elsewhere in the country. On average, respondents had more than nine years of completed formal education, and nearly 14 percent of the control sample was enrolled in some form of educational program at baseline.[17] Except for age and reported motherhood, table 1 does not show any systematic differences across a wide range of observable characteristics between the treatment and control groups, which suggests that random allocation of study participants worked well. An F-test cannot reject the joint hypothesis that each coefficient obtained from a regression of treatment on all variables listed in table 1 is equal to 0. To attenuate the effect of potential biases and improve precision, the educational and demographic variables are used as controls in the estimating equation.

To compare the study participants to a nationally representative sample of urban job seekers, this paper also uses data from the Urban Employment Unemployment Survey (UEUS), which was conducted at about the same time as our baseline survey. Column 1 of table S1.2 in the supplementary online appendix is based on the complete UEUS sample collected from across all urban areas in Ethiopia. Column 2 restricts the sample to Addis Ababa, and column 3 restricts this further to those who had the age and education profile that match our own survey sample. This accounts for approximately 20 percent of the Addis Ababa sample in the UEUS. While there are some differences between the study sample and the nationally representative sample of job seekers, the differences are hardly economically meaningful.

3.3. Empirical Strategy

There is considerable self-selection at various stages during the hiring and onboarding process for factory work. Not all eligible applicants in the treatment group participated in the job interviews, not all applicants who passed the interview successfully took the job, and not all applicants who started working in the factories stayed on the job until follow-up data was collected. Table S1.3 in the supplementary online

16 Lower and upper bounds are estimated using a method of bounding developed by Lee (2009) on employability and earnings at the first follow-up. The study finds that the results are not sensitive to the bounding. Therefore, this paper refrains from showing these results formally and does not discuss these results further.

17 This typically involves enrollment in evening and weekend classes in public schools.

where y_{it} is an outcome measured for respondent i at the eight-month ($t = 1$) or four-year ($t = 2$) follow-up. The variable $treat_i$ is an indicator equal to 1 if the respondent was assigned to the treatment group and 0 otherwise. X_{i0} includes a set of control variables measured at baseline including age, a dummy for marital status, a dummy for motherhood, a dummy for migration status, the highest grade completed, and evaluation duration, measured as number of days between baseline and follow-up surveys. The specifications also include month-of-survey dummies to control for the possibility of COVID-19 impacts confounding the results.[18] y_{i0} is the outcome measured at baseline. The results section's focus is on β_2, which measures the treatment impact, and does not report β_1 or β_3. To account for attrition, each observation is weighted by the inverse of its predicted probability of continuing in the study. All reported control group means are also adjusted using these weights.

This study tests whether treatment had any impact on a wide range of outcome variables measured in the two follow-ups. For each family of outcome variables, sharpened q-values are constructed to control for multiple hypothesis testing (Benjamini, Krieger. and Yekutieli 2006). Sharpened q-values are designed to control for false discovery rates when simultaneously testing for an effect of treatment on several set of outcome variables that may be correlated. As a further robustness test, a standardized index is created for each family of outcomes, and the treatment results on these aggregates are presented in table S1.4 in the supplementary online appendix.

Aside from the ANCOVA specification, the results from a simple mean comparison of the outcome y_{it} between the treatment and control groups for the two follow-ups (labeled "ITT difference" in the tables) are shown. The sharpened q-values for these results are also computed.

4. Results

The results are split into short-run (eight-month) and long-run (four-year) impacts, which are summarized in tables 2 through 4. The set of outcomes considered includes variables that capture employment, income, expenditures, health, and expectations and perceptions about factory work.

4.1. Impacts on Employment

The immediate objective of the facilitation intervention was to ease the logistical and procedural hurdles during the factory job application process. Table 2 shows that the likelihood of having had a factory job is more than 50 percent greater than for the control group in the short run, indicating that the job-facilitation intervention was successful in bridging logistical constraints on seeking employment in the industrial park. More importantly, the intervention was successful in affecting the employment status eight months after the treatment. Table 2 indicates that the treatment group is nearly 25 percent more likely to be employed, largely driven by wage, particularly factory work. The amount of time that individuals were employed in the six months leading up to the first follow-up survey also increases.

Consistent with the hypothesis that industrial work offers steadier hours, table 2 shows that the intervention is associated with roughly five more hours of work per week in the treatment sample. The impact on self-employment is close to zero, suggesting that the factory jobs represented an addition to the set of job opportunities available to the target group and did not merely crowd out other income-generating activities in the short run.

The results from the four-year follow-up indicate that the higher levels of employment in factory work observed in the short run persist but that overall employment effects dissipate. The last two columns

18 The four-year follow-up survey was conducted between February and April 2020. The first COVID case in Ethiopia was recorded on March 13, 2020, and a state of emergency outlining mitigation strategies including mobility restrictions and social distancing was introduced on April 8, 2020. More than 93 percent of the second follow-up survey was completed before the state of emergency was declared.

Table 2. Impacts on Employment

Dependent variables	2017			2020		
	Control mean (1)	ITT difference (2)	ITT ANCOVA (3)	Control mean (4)	ITT difference (5)	ITT ANCOVA (6)
Ever employed in factory work (yes = 1)	0.280	0.178***	0.181***	0.406	0.115***	0.099**
	[0.450]	(0.038)	(0.038)	[0.492]	(0.043)	(0.040)
		{0.001}***	{0.001}***		{0.05}**	{0.107}
Currently employed (yes = 1)	0.466	0.117***	0.102**	0.589	−0.016	−0.024
	[0.499]	(0.043)	(0.043)	[0.493]	(0.042)	(0.042)
		{0.01}**	{0.027}**		{1.0}	{1.0}
Currently employed in self-employment (yes = 1)	0.037	0.014	0.013	0.117	−0.015	−0.026
	[0.190]	(0.017)	(0.017)	[0.322]	(0.027)	(0.028)
		{0.062}*	{0.086}*		{1.0}	{1.0}
Currently employed in wage employment (yes = 1)	0.434	0.106**	0.092**	0.492	0.001	−0.002
	[0.497]	(0.043)	(0.044)	[0.501]	(0.043)	(0.042)
		{0.013}***	{0.029}**		{1.0}	{1.0}
Currently employed in factory work (yes = 1)	0.102	0.092***	0.087***	0.064	0.051**	0.039*
	[0.303]	(0.028)	(0.029)	[0.245]	(0.023)	(0.023)
		{0.004}***	{0.008}**		{0.083}*	{0.357}
Time employed in the past 6 months (in months)	2.913	0.625***	0.529**	3.265	−0.054	−0.087
	[2.735]	(0.232)	(0.225)	[2.806]	(0.240)	(0.243)
		{0.01}**	{0.026}**		{1.0}	{1.0}
Hours worked in 7 days	23.786	5.243**	4.725**	66.364	0.487	−0.838
	[25.785]	(2.213)	(2.236)	[70.591]	(6.057)	(5.940)
		{0.013}**	{0.031}**		{1.0}	{1.0}

Source: Authors' analysis based on data from the baseline, first follow-up, and second follow-up surveys.
Note: The table reports the intent-to-treat (ITT) estimates of the effect of the job-facilitation intervention on key outcomes related to employment and job search. The sample constitutes 687 respondents who were interviewed in all three waves. The ANCOVA regression specifications use controls for individual characteristics, such as age, marital status, schooling, time between baseline and follow-up, month survey dummies, and *woreda* fixed effects. To account for attrition, all observations are weighted by the inverse of their predicted probability of being tracked at follow-up. Standard deviations in brackets. Standard errors in parenthesis and q-values in curly brackets are reported underneath each estimated coefficient; *q-values* are calculated following the sharpened procedure proposed by Benjamini, Krieger, and Yekutieli (2006). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

of table 2 show that treatment individuals are no more likely to have higher rates of current employment, wage employment, or self-employment than those in the control group. As indicated in the last two rows, the number of months employed in the six months before the second follow-up survey and weekly hours of work are similar between the two groups as well. The long-run results are mostly driven by the fact that the control group caught up to the treatment group by finding alternative employment opportunities. Four years after the intervention, average weekly labor hours converged—67 hours for the treatment and 66 hours for the control individuals.

4.2. Impact on Earnings, Expenditures, and Savings

In the short run, helping job applicants obtain factory work significantly increases their income (first row of table 3). Earnings received in the last four weeks were about 200 birr higher in the treatment group, which is an increase of nearly 33 percent over the mean of the control group at baseline. Consistent with the higher income that the treatment group earned, table 3 shows that the intervention is associated with larger nonfood expenditures and higher savings in the treatment group. These findings contrast with a key result in Blattman and Dercon (2018), who do not find significant income effects from offering factory jobs. These results are also not artifacts of multiple hypotheses testing as shown by low q-values throughout the table.

Table 3. Impacts on Income, Expenditures and Savings

	2017			2020		
Dependent variables	**Control mean** (1)	**ITT difference** (2)	**ITT ANCOVA** (3)	**Control mean** (4)	**ITT difference** (5)	**ITT ANCOVA** (6)
Total cash earnings in the past 4 weeks (in birr)	610 [686]	210.886*** (61.830) {0.003}***	197.631*** (62.115) {0.005}***	1334 [1360]	121.215 (163.797) {0.738}	125.911 (173.663) {1.00}
Food expenditure in past 7 days (in birr)	127 [130]	5.111 (11.335) {0.195}	12.872 (9.896) {0.107}	384 [344]	−31.903 (28.466) {0.289}	−20.969 (28.868) {1.00}
Nonfood expenditure in past month (in birr)	335 [349]	55.852* (30.490) {0.069}*	55.408* (30.443) {0.054}*	1109 [1174]	−70.117 (96.942) {0.738}	−41.019 (93.079) {1.00}
Savings from earnings in past 4 weeks (in birr)	61.0 [173.0]	35.211** (16.412) {0.051}*	34.944** (15.950) {0.040}**	316 [788]	−31.625 (63.608) {0.738}	−37.526 (61.057) {1.00}

Source: Authors' analysis based on data from the baseline, first follow-up, and second follow-up surveys.

Note: The table reports the intent-to-treat (ITT) estimates of the effect of the job-facilitation intervention on key outcomes related to income and expenditure. Columns 1 and 4 present the control means associated with each of the follow-up surveys separately. The sample constitutes 687 respondents who were interviewed in all three waves. The ANCOVA regression specifications use controls for individual characteristics, such as age, marital status, schooling, time between baseline and follow-up, month survey dummies, and *woreda* fixed effects. To account for attrition, all observations are weighted by the inverse of their predicted probability of being tracked at follow-up. Standard deviations in brackets. Standard errors in parenthesis and q-values in curly brackets are reported underneath each estimated coefficient; *q-values* are calculated following the sharpened procedure proposed by Benjamini et al. (2006). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

That said, four years after the intervention, the study finds no statistically significant treatment impacts on earnings, expenditures, or savings. The earnings gains that the treatment individuals enjoyed over the control group at the first follow-up are erased by the four-year follow-up (fig. S1.1). The reason for this may be that the employment rate of the two groups converges in the long run (see table 2). Given the substantive entry-level pay difference between factory work and alternative jobs observed in the short run, the lack of long-run earning impacts may result from infrequent wage adjustment or slow wage growth in industrial parks.

4.3. Impacts on Health

Industrial work can potentially expose workers to health risks. In the short run, the study finds that a standardized strenuosity score, which measures difficulty with daily (and work-related) activities, increases for the treated group (table 4).[19] Respondents were also asked to assess the healthiness of factory jobs.[20] The estimates presented in the second row of table 4 suggest that the treatment group is more likely to consider factory work unhealthy. These results mirror the adverse health impacts that Blattman

19 This score relies on 10 questions to assess individuals' capability to undertake activities of daily life (ADLs). Based on individuals' assessment of their capacity. More precisely, the questions ask respondents to rate whether following activities can be done "easily," "with slight difficulty," "with great difficulty" or if the respondent is "unable to do it": a scale 1–4. The score is the sum of the answers. The ADLs are: "able to walk for 2 kilometers," "able to carry a 20-liter container of water for 20 meters," "able to carry out your usual daily activities," "to work on your feet outdoors for a full day," "able to stand at a workbench or assembly line for 6 to 8 hours," "able to stand up after sitting down on a chair," "able to stand up after sitting down on the floor," "able to bend, kneel or stoop," "able to wipe the floor," and "able to read a book at an arm's length."

20 This measure is based on the question "How would you rate the healthiness of a garment factory job?" The three available options are "healthy," "somewhat health," and "not healthy." Factory work considered as unhealthy if the respondents chose "not healthy."

Table 4. Impacts on Health, Expectations, and Perceptions

Dependent variables	2017			2020		
	Control mean (1)	ITT difference (2)	ITT ANCOVA (3)	Control mean (4)	ITT difference (5)	ITT ANCOVA (6)
Health						
Standardized strenuosity score	−0.157	0.199**	0.188**	−0.025	0.015	−0.004
	(0.890)	(0.080)	(0.082)	0.894	(0.078)	(0.074)
		{0.007}***	{0.01}**		{1.00}	{1.00}
Consider garment factory jobs as unhealthy (yes = 1)	0.563	0.175***	0.160***	0.865	0.004	0.024
	[0.497]	(0.041)	(0.042)	[0.343]	(0.030)	(0.028)
		{0.001}***	{0.001}***		{1.00}	{1.00}
Perception and Expectations						
Subjective factory job quality (scale 1–10)	6.079	−1.380***	−1.297***	4.847	0.112	0.218
	[2.11]	(0.185)	(0.190)	[7.566]	(0.540)	(0.640)
		{0.001}***	{0.001}***		{0.264}	{0.225}
Expected monthly earnings from working at garment factory (in birr)	1558	−121.372***	−97.864**	1500	−149.98**	−122.851*
	[444]	(39.608)	(39.833)	[891]	(69.677)	(65.529)
		{0.004}***	{0.025}**		{0.070}*	{0.102}
Perceives factory jobs provide a steady income (yes = 1)	0.616	0.010	0.024	0.579	−0.091**	−0.094**
	[0.479]	(0.042)	(0.042)	[0.495]	(0.043)	(0.041)
		{0.252}	{0.182}		{0.070}*	{0.100}
Perceives factory jobs as permanent (yes = 1)	0.931	−0.052**	−0.039	0.589	−0.069	−0.069*
	[0.254]	(0.024)	(0.024)	[0.493]	(0.043)	(0.041)
		{0.020}**	{0.071}*		{0.076}*	{0.105}

Source: Authors' analysis based on data from the baseline, first follow-up, and second follow-up surveys.

Note: The table reports the intent-to-treat (ITT) estimates of the effect of the job-facilitation intervention on individual's health, expectation and perceptions of factory jobs. The standardized strenuosity score is based on how well the respondents can carry out 10 basic activities. These activities ask whether the respondent is able to walk, carry, stand, bend, read, kneel or stoop under varying conditions. The respondent is asked to indicate if she can do the activities easily, with slight difficulty, with great difficulty or is unable to do it at all. Expected earnings are drawn from the question "How much do you think you can earn per month as a worker in a garment factory job." Columns 1 and 4 present the control means associated with each of the follow-up surveys separately. The sample constitutes 687 respondents who were interviewed in all three waves. The ANCOVA regression specifications use controls for individual characteristics, such as age, marital status, schooling, time between baseline and follow-up, month survey dummies and *woreda* fixed effects. To account for attrition, all observations are weighted by the inverse of their predicted probability of being tracked at follow-up. Standard deviations in brackets. Standard errors in parenthesis and *q*-values in curly brackets are reported underneath each estimated coefficient; *q-values* are calculated following the sharpened procedure proposed by Benjamini, Yekutieli (2006). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

and Dercon (2018) reported.[21] The radar graph in fig. S1.2 decomposes the index and plots impacts and corresponding confidence intervals for each of the activities in isolation. Perhaps unsurprisingly, the two activities most closely associated with and required for factory work (working on feet and standing at work bench) are driving the results on the strenuosity score.

Four years after the intervention, the adverse health impacts disappear, which may be explained by the following. First, women experiencing negative impacts in the short run may have left factory employment. Second, the negative health impacts may be temporary. There is evidence corroborating the second explanation. The study finds that the likelihood of current factory employment at four-year follow-up is not correlated with reported health status at the eight-month follow-up. Table 2 showed that individuals in the treatment group are more likely to be employed in factories, and table 4 indicates that their current jobs are no more physically demanding or unhealthy than those of the control group. Figure S1.3 also presents the same decomposition of the health index using the radar graph for long-run impacts. The graph supports the conclusion that health problems uncovered in the short run did not persist over time.

21 In a recently published five-year follow-up study, Blattman, Dercon and Franklin (2022) find that the adverse health effects observed after one year of follow-up dissipate over time.

This result is consistent with that of Blattman, Dercon, and Franklin (2022), who also find that negative health impacts detected one year after the intervention decay over time.

4.4. Impacts on Expectations and Perceptions

One possible explanation for the high rate of worker turnover and the short duration of workers' spells in factories is an ex-ante mismatch between applicants' job-related expectations and the reality of factory work. Recent qualitative studies that explore worker and manager interactions in the textile, garment, and leather manufacturing industries find that expectation mismatch is a key driver of worker turnover in Ethiopia (Barrett and Baumann-Pauly 2019; Hardy and Hauge 2019). The present study tests whether treatment group participants adjusted their expectations regarding factory work after having experienced this type of work.

The results are shown in the lower panel of table 4 and suggest that treatment group participants lowered their expectations regarding the earning potential of factory work and adjusted their expectations downward even more in the long run. They were also less likely to view factory work as a permanent job. Again, this decline is more pronounced in the long run. Although treatment applicants had significantly more negative perceptions of the job quality and healthiness of factory work in the short run, their views do not differ from those in the control group in the long run, partly because the control group also developed more negative views about factory work themselves. This finding is plausible if one accepts that information about factory jobs spreads, for example, through social networks or public media over time.[22]

Intuitively, these findings are not surprising. The short-run health impacts that workers experience are likely to make factory work less preferable compared to alternatives in the long term (Blattman and Dercon 2018). By the very nature of the organizational hierarchy in factories, upward mobility of young workers' careers in the garment industry is limited. This industry employs many relatively poorly educated, low-skilled workers who typically remain at the bottom of the organizational pyramid. Even with tenure, the chances of career progression to positions beyond line management or supervisory levels are minimal. The recruitment pool for these positions differs, and college graduates from the outside commonly fill openings. Further, employment alternatives, such as the prospect of being employed as a domestic worker in the Gulf or the Middle East, are attractive outside options (Hardy and Hauge 2019).[23] Female workers face demands on their time related to childcare and family responsibilities, as well as potentially constraining social norms, once they marry, that can effectively end careers in the garment industry. Therefore, many workers will consider jobs in labor-intensive manufacturing industries as temporary or at least nonpermanent.

4.5. Discussion

Tying together these results, it seems evident that factory jobs in the present context are not attractive enough to convince large numbers of job seekers to interview for these positions. Additionally, even when applicants accept factory positions, workers are less likely to stay on the job for a longer period of time, reducing the potential for a job-facilitation intervention to have lasting effects in the long run. This finding resonates well with previous work on this topic and serves as a reminder that worker turnover can generate a misallocation of talent and lead to welfare loss. In particular, those who quit early may not have had sufficient time to learn about important aspects of the job and the job's match with their own preferences,

22 The control mean associated with job quality indicators and the perception index decline substantially between 2017 and 2020 (table 4).

23 Blattman and Dercon (2018) find that 5.3 percent of their sample emigrated to the Middle East and that the likelihood of migration is higher in the two treatment arms they studied. Anecdotally, factory job income provides the means through which savings can be accumulated, which are necessary to finance young women's transitions to the Gulf or the Middle East.

4.6. Robustness Checks

Multiple hypothesis testing is a possible concern when testing numerous outcomes at the same time. In addition to the sharpened q-values the paper reports in the main results tables, this concern is addressed by aggregating related outcome variables into four distinct index families. Each index is then transformed into a z-score to generate standardized indices, one for each family of outcomes.[25] Following this procedure, all the outcome variables presented in tables 2, 3, and 4 are combined to generate indices on employment, income and spending, health, and expectations and perceptions. Table S1.4 reports the simple ITT differences and ANCOVA treatment effects for the indices. The intervention affects each of the four indices, but these effects are observed only in the short run. There are no long-run effects with the notable exception of the expectation and perception index.

The level of attrition in the balanced sample is another concern that is addressed in different ways. First, the study applies inverse probability weighting throughout for the main results. Second, the short-run results are re-computed using all observations available without restricting the sample to the full panel respondents only. That is, information from all 827 baseline respondents who were interviewed in the first follow-up is used. In doing so, the results are qualitatively the same as the impacts reported in the main regression tables, and the estimated coefficients are very close in magnitude. Only the ITT estimate on savings is no longer significant, but the adverse effect on the perception of factory jobs as permanent employment now is. Finally, a series of bounding approaches are implemented to check the sensitivity of the results to different assumptions on the nature of missing data due to attrition. This mainly relies on two approaches and focuses on the main results—impacts on employability and earnings at the first follow-up. For employment-related indicators, the method of bounding developed by Lee (2009) is used. For earnings, missing data is imputed using a similar procedure to what is commonly done in the related literature (Abebe et al. 2021; Blattman, Fiala, and Martinez 2014; Karlan and Vadivia 2011). This method offers a test of the sensitivity of the results to potential systematic attrition, where attritors in the control and treatment groups are assumed to exhibit different patterns that lead to overestimation (lower bounds) or underestimation (upper bounds) of treatment effects. The bounding results suggest that the main results are robust to attrition effects. These results are not reported, but they are available upon request.

5. Conclusion

New economic opportunities such as factory jobs in a growing manufacturing sector are thought to provide a potential avenue out of poverty for many people in economies undergoing structural transformation. This study facilitated access to such factory jobs at the Bole Lemi industrial park on the outskirts of Addis Ababa to study their welfare impacts. To do so, a randomly selected sample of female job seekers are provided logistical support during the hiring process to help them overcome entry barriers to formal employment that women often face. These barriers tend to be especially constraining for those with weak networks and limited labor market experience, which often characterizes young women at the beginning of their professional careers, in particular if they are migrants. The study estimates the impact of the intervention eight months and four years after the initial job screening and application phase.

The results indicate that, in the short run, a light-touch job-facilitation intervention can have large impacts on applicants' success rates in finding employment and in improving earnings, although adverse health impacts accompany these effects. In the long term, applicants in the control group seemingly catch up and find other employment opportunities, which erases the earnings advantage of those in the treatment group but also cancels out the adverse health impacts. Moreover, negative perceptions of factory work increase in both groups. Part of this is likely due to information updating as more potential work-

25 In case of continuous variables, a dummy is generated that takes 1 if its value is larger than the median of the variable or 0 otherwise.

ers learn about the peculiarities of factory work. However, the absence of sustained growth in earnings raises questions about long-term career prospects for workers and about talent acquisition, retention, and productivity for firms, which have important implications for long-term industrial development.

Data Availability Statement

The data underlying this article are published in the World Bank's microdata library https://microdata.worldbank.org/index.php/catalog/6199. Replication files can be accessed at https://github.com/girumabe/Factory-Jobs.

Conflict of Interest

The authors declare that there is no conflict of interest related to this research work. The authors do not stand to benefit, either financially or professionally, from reporting either null, positive or negative impacts from the experiment in this research.

References

Abebe, G., A.S. Caria, M. Fafchamps, P. Falco, S. Franklin, and S. Quinn. 2021. "Anonymity or Distance? Job Search and Labour Market Exclusion in a Growing African City." *Review of Economic Studies* 88(3): 1279–310.

Abebe, G., A.S. Caria, and E. Ortiz-Ospina. 2021. "The Selection of Talent: Experimental and Structural Evidence from Ethiopia." *American Economic Review* 111(6): 1757–806.

Abel, M., R. Burger, E. Carranza, and P. Piraino. 2019. "Bridging the Intention-Behavior Gap? The Effect of Plan-Making Prompts on Job Search and Employment." *American Economic Journal: Applied Economics* 11(2): 284–301.

Animashaun, J., and L. Emediegwu. 2023. "Structural Transformation and Gender-Specific Labour Market Frictions: Evidence from Nigeria." Working paper WP065. Structural Transformation and Economic Growth (STEG), London, United Kindom.

Ashraf, N., O. Bandiera, E. Davenport, and S.S. Lee. 2020. "Losing Prosociality in the Quest for Talent? Sorting, Selection, and Productivity in the Delivery of Public Services." *American Economic Review* 110(5): 1355–94.

Banerjee, A.V., and S. Sequeira. 2020. "Spatial mismatches and imperfect information in the job search." Discussion Paper No. DP14414. Center for Economic Policy Research. Washington, DC, USA. https://ssrn.com/abstract=3547347

Barrett, P., and D. Baumann-Pauly. 2019. "*Made in Ethiopia: Challenges in the Garment Industry's New Frontier*." Stern Center for Business and Human Rights. New York University. New York, NY USA.

Benjamini, Y., M.A. Krieger, and D. Yekutieli. 2006. "Adaptive Linear Step-Up Procedures That Control the False Discovery Rate." *Biometrika* 93(3): 491–507.

Berik, G., Y.V.D.M. Rodgers, and J.E. Zveglich. 2004. "International Trade and Gender Wage Discrimination: Evidence from East Asia." *Review of Development Economics* 8(2): 237–54.

Bernard, A.B., and J.B. Jensen. 1999. "Exceptional Exporter Performance: Cause, Effect, or Both?" *Journal of International Economics* 47(1): 1–25.

Bezawagaw, G., N. Dihel, M. Geiger, and Z. Kelbore. 2018. "Ethiopia Economic Update: The Inescapable Manufacturing Services Nexus: Exploring the Potential of Distribution Services." World Bank Group, Washington, DC, USA. http://documents.worldbank.org/curated/en/811791526447120021/The-inescapable-manufacturing-services-nexus-exploring-the-potentialof-distribution-services

Blattman, C., and S. Dercon. 2018. "The Impacts of Industrial and Entrepreneurial Work on Income and Health: Experimental Evidence from Ethiopia." *American Economic Journal: Applied Economics* 10(3): 1–38.

Blattman, C., S. Dercon, and S. Franklin. 2022. "Impacts of Industrial and Entrepreneurial Jobs on Youth: 5-Year Experimental Evidence on Factory Job Offers and Cash Grants in Ethiopia." *Journal of Development Economics* 156:102807.

WPS4823

Policy Research Working Paper 4823

Why Do People Die in Earthquakes?

The Costs, Benefits and Institutions of Disaster Risk Reduction in Developing Countries

Charles Kenny

The World Bank
Sustainable Development Network
Finance Economics & Urban Department
January 2009

Policy Research Working Paper 4823

Abstract

Every year, around 60,000 people die worldwide in natural disasters. The majority of the deaths are caused by building collapse in earthquakes, and the great majority occurs in the developing world. This is despite the fact that engineering solutions exist that can almost completely eliminate the risk of such deaths. Why is this? The engineering solutions are both expensive and technically demanding, so that the benefit-cost ratio of such solutions is often unfavorable compared with other interventions designed to save lives in developing countries. Nonetheless, a range of public disaster risk-reduction interventions (including construction activities) are highly cost effective. The fact that such interventions often remain unimplemented or ineffectively executed points to a role for issues of political economy. Building regulations in developing countries appear to have limited impact in many cases, perhaps because of limited capacity and the impact of corruption. Public construction is often of low quality—perhaps for similar reasons. This suggests approaches that emphasize simple and limited disaster risk regulation covering only the most at-risk structures and that (preferably) can be monitored by non-experts. It also suggests a range of transparency and oversight mechanisms for public construction projects.

This paper—a product of the Finance Economics & Urban Department, Sustainable Development Network—is part of a larger effort in the department to analyze the economics of natural disasters. This paper is prepared as a background paper to the joint World Bank-UN Assessment of the Economics of Disaster Risk Reduction. Funding from the Global Facility for Disaster Reduction and Recovery is gratefully acknowledged. Policy Research Working Papers are also posted on the Web at http://econ.worldbank.org. The author may be contacted at ckenny@worldbank.org.

Introduction

Natural disasters have killed thousands of people in the space of a few minutes. Their after-effects can kill thousands more over subsequent weeks and years. They can have a huge cost in terms of damaged buildings and infrastructure. And yet, many of the deaths and much of the damage is preventable. This paper focuses on where the great majority of the human cost of disasters is concentrated – the developing world. It examines the reasons behind continued, wide-scale death and destruction from disasters, and what these reasons suggest about effective responses in the context of developing countries.

Earthquakes kill thousands every year. The 8.0 magnitude Sichuan Earthquake in 2008 killed over 70,000 people. The 6.8 earthquake in Algeria in 2003 killed 2,700, the 1999 Marmara earthquake in Turkey killed 17,000 (Escaleras, Anbarci and Register, 2007, Anbarci, Escaleras and Register, 2005). Earthquakes account for the majority of deaths from a range of natural disasters which amounts to about 60,000 people a year worldwide – around 90 percent of which occur in developing countries (OECD, 2008).

Most earthquake deaths are related to building collapse or damage. In the 2008 Sichuan earthquake, for example, hundreds of thousands of buildings – including numerous public buildings such as schools and hospitals – collapsed. Beyond the human toll, the cost of this physical destruction can be considerable. The Marmara earthquake was estimated to have had a direct economic impact of over $5 billion (World Bank, 2005).[2]

It is worth noting that while the majority of the human toll of disasters is borne by developing countries, most property damage occurs in wealthy countries (Box 1). The high human toll in developing countries is because despite the fact that it is well known how to design buildings to survive an earthquake well enough to avoid collapse, buildings in poor countries are usually not constructed according to such designs (Box 2). Correct construction of large, reinforced concrete structures in particular takes skills, expertise, time and money. All too often the capacities or the incentives required to ensure robust construction has been missing.

Fragile construction, alongside higher-risk land use practices, account for much higher death rates in similar-sized earthquakes close to population centers in the developing as opposed to the developed world. The 1988 earthquake in Armenia had half the energy release of the 1989 earthquake in Lolma Prieta near San Francisco, California, and yet caused 25,000 deaths compared to 100 in San Francisco.[3] The 2003 Paso Robles quake in California had the same power as the Bam quake in Iran, the death toll was two in California and 41,000 in Bam. According to later engineering analysis, a major

[2] To provide some indication of the human toll of an earthquake in a developing country setting as measured in economic terms, using a value of statistical life set at $250,000, the loss of life alone in the Marmara earthquake 'cost' $4.25 billion. This estimate comes with all of the considerable caveats about putting a financial value on human life.

[3] The nature of construction in the two countries –with a much greater use of wood in San Francisco—may also have played a considerable role. Having said that, wooden buildings in the Kobe earthquake were some of the most likely to collapse (Ghosh, 1995)

difference was strict adherence to tough zoning and building codes in California compared to Iran and Armenia (see Box 3).

Box 1: Lives versus property: the differential toll of disasters in countries rich and poor

There is *absolutely no overlap* between the top twenty most costly insured disasters worldwide 1970-2005 and the top twenty worst catastrophes in terms of lives lost. All of the most costly events in terms of property damage hit wealthy countries (where the density of economic activity is higher), all of the most deadly hit developing countries. In addition, eighteen out of twenty of the most costly were hurricanes, typhoons and storms, eleven out of twenty of the most deadly were earthquakes (from OECD, 2008). Why is this? Earthquake deaths in particular can be prevented by engineering approaches that are widely applied in wealthy countries but rare in the developing world. It is more difficult to comprehensively prevent flood and wind damage using similar approaches.

Box 2: Poorly constructed buildings collapse in earthquakes

In the 2008 Sichuan earthquake, early evidence suggests that non-ductile and un-reinforced concrete buildings were particularly prone to collapse. This has been a ubiquitous problem in the developing World. Jain (2005) suggests that in 1999-2000 over 6,000 school buildings were constructed in Gujarat, India using seismically weak pre-cast construction technology. Three quarters of these schools were seriously damaged or collapsed during the 2001 Bhuj earthquake. Deaths from the 1999 Marmara earthquake in Turkey were blamed on collapse due to poorly-constructed reinforced concrete frames, construction using concrete diluted with too much sand, and construction near fault-lines (Escaleras, Anbarci and Register, 2007, Anbarci, Escaleras and Register, 2005). The Turkey experience illustrates that, while well-constructed reinforced concrete buildings are less likely to collapse in earthquakes than un-reinforced masonry, construction is technically complex and requires skilled workers motivated to meet high standards. And when such building do collapse because they are poorly built, they are more likely to kill occupants.

It is not that the world lacks the knowledge to considerably reduce the number of fatalities from natural disasters, then. A combination of better land use, better construction practices and improved preparedness would dramatically reduce risk across the full range of disasters. Why, then, are these mitigation measures not adopted in the developing world?

Disaster protection may be a luxury good in economic terms – demand for protection rises more than proportionally as incomes increase. In poor countries such protection may provide lower or more uncertain returns than a number of other investments in wellbeing – the vast majority of people do not die in disasters, and it may be easier and cheaper to prevent other causes of death. Construction of disaster-resistant buildings might be technically complex, taxing limited design and construction capacities in developing countries. The knowledge and understanding of citizens regarding the risks and mitigation measures around disasters may be limited, reducing demand for safe construction and/or the ability to ensure such construction is being carried out. The design and enforcement of regulations to ensure good construction may also be complex and prone to failures of governance including corruption. We will see that all of these factors may play a role in the outcome of greater natural disaster fatalities in developing countries. In turn, this suggests a number of recommendations for countries regarding construction practices, public mitigation measures and regulation.

Box 3: Less damage is done by disasters in rich countries because of code enforcement

It is a repeated finding in the literature that fatalities resulting from an earthquake are inversely proportional to the level of per capita income (Anbarci, Escaleras and Register, 2005). This is in large part because of the widespread adoption of proactive measures to reduce disaster mortality introduced in developed countries since the Second World War. As codes improved in the developed world building collapse during earthquakes became less frequent. In Kobe, buildings constructed after a 1981 revision of Japan's building codes were far less likely to collapse in the 1995 earthquake than older buildings. And all of the reinforced concrete buildings which saw story collapse (particularly likely to kill occupants) were built prior to a 1971 code change regarding beam and column ductility (Ghosh, 1995). Similarly, Wharton (2008) notes that within the US, houses in one county hit by hurricane Charley in 2004 saw insurance claim frequencies that varied dramatically depending on if they had been built before or after 1996, when new wind-resistant standards were introduced. Homes built after 1996 had a 60 percent lower claim frequency than those built before 1996.

Individuals, Unprompted, Do Little to Prepare for or Insure Against Natural Disasters

Worldwide, absent legal requirements, individuals are unlikely to take even fairly simple measures to reduce disaster risk. In South Florida, where hurricanes are frequent, it appears that many residents fail to take short-term preparatory measures such as securing bottled water and filling their cars up with gas when storms threaten. Only 17 percent of people surveyed living along the Atlantic and Gulf coasts in the US suggested that they had taken steps to storm-proof their home in 2006 (Wharton, 2008). In 1974, a survey of California homeowners in earthquake-prone areas found that only 12 percent of respondents had adopted protective measures against earthquakes, and the figure was, if anything, lower by 1989. Similarly, even after the Marmara earthquake, willingness of householders to pay for earthquake-proof retrofitting in Turkey, even using subsidized credit, was low (World Bank, 2005).[4]

The market value attached to risk-reduction measures also appears to be limited. While it appears that housing rents in Tokyo are lower in areas of high risk of earthquake damage, and that rents for buildings in high-risk areas constructed prior to strengthened earthquake codes are lower still, the evidence suggests that retrofitting would still only make financial sense for building owners because of a considerable government subsidy (Nakagawa, Saito and Yamaga, 2007). (Furthermore, given that high risk areas are on historically swampy ground traditionally home to the poor, and older buildings are likely

[4] Even those who do respond to calls for preparedness unsurprisingly favor preparations that are easier to make (moving furniture) over more time-consuming and expensive measures such as structural alteration (Mileti et. al., 1992).

Box 5: Does lack of information drive lack of response?

In the US, information campaigns regarding disaster risk have apparently had little impact on behavior (Praeter and Lindell, 2000). In California, a 1971 law mandating that prospective buyers be informed of a property's location in a surface fault rupture zone had no impact on property prices in those zones (Palm, 1981). In Japan, Shaw et. al. (2004) find that formal education plays a limited role in promoting actions towards earthquake preparedness. A similar conclusion applies to the value of risk-reduction regarding environmental health risk in the US. A study comparing housing prices in areas that qualified for Superfund cleanups compared to areas that narrowly missed qualification found that being put on a list that suggests the need for a significant environmental cleanup effort because of health risks does not reduce rental prices and that cleanups themselves are associated with statistically insignificant changes in property values, rental rates and population (Greenstone and Gallagher, 2008).

The Benefit-Cost of Disaster Mitigation

If neither moral hazard, a lack of recognition of risks or a simplistic valuation of prevention measures as a 'luxury good' appears to fully account for limited mitigation effort on the side of individuals and frequently limited demand for public mitigation measures, might there be a role for the rational calculation of costs and benefits in explaining inaction?

Calculation of the benefits and costs of disaster risk reduction measures such as building strengthening/retrofit involves estimates and assumptions covering a range of factors – strengthening/retrofit costs, building replacement costs, the risk of a natural disaster (and of the scale of that disaster), the risk of damage if a natural disaster does occur, the cost of that damage in both financial and human terms and the discount rate. These last estimates, in particular, involve a difficult calculus regarding the economic value of a human life as well as how differently we value that life today over a life 30 years hence.

Take a base case scenario using estimates many of which are drawn from an exercise comparing costs and benefits of retrofitting apartment buildings in Istanbul to make them earthquake-resistant. This involves an apartment building with a $250,000 reconstruction cost and $80,000 retrofit/strengthened construction cost (that will prevent any damage in an earthquake). The probability of an earthquake in any year is set at 2%, and the chance that the building will collapse in an earthquake is 10%. If it does collapse, ten people are assumed to die, with a statistical value of life set at $250,000. The discount rate is set at 5 percent and the span of years over which we account for costs and benefits is 30.[10] Under this scenario, the benefits of retrofitting about equal costs – the benefit-cost ratio is 1.0.

[10] The construction, retrofitting, deaths are taken from Smyth et. al. based on Istanbul data. Ther value of a statistical life is set at $250,000 for reasons that will be clear in the text –as opposed to the Smyth et. al. figure of $1,000,000.

Unsurprisingly, changing the base case assumptions (sequentially) has a significant effect on benefit-cost ratios (see Table 1). Halving the retrofit/improved construction cost, doubling the chance of an earthquake or the chance of collapse or the number or value of deaths in a collapse all approximately double the ratio of benefits to retrofitting over costs. Halving the discount rate of doubling the years accounted has a smaller, but still significantly positive impact on the benefit-cost ratio. The effects are similar in magnitude and opposite in direction under the reverse adjustments.

But a significant problem with calculating the net benefit of retrofitting is that all of these numbers are open to significant uncertainty, variation and debate – the base case is very uncertain. Retrofitting costs will vary significantly by building type and location. The probability of a large earthquake strong enough to pose significant risk to buildings is of course highly location-specific and open to considerable uncertainty. How likely a particular un-retrofitted apartment building is to collapse is similarly dependent on a range of design and location factors, some known, some unknown. The number and statistical value placed on people who die in a collapse, which drives the benefit calculation far more than the cost of building replacement in developing country settings, is both uncertain and controversial. Similarly, the appropriate discount rate to use is a matter of some debate, and may vary with regard to the individual and the government.

Looking first at strengthening/retrofit costs, initial construction using designs that are less likely to collapse in earthquakes adds to building cost. One recent estimate for school buildings in a developing country is around 8 percent (Bahatia 2008a) more advanced protection for at-risk housing in Box 7 suggests a similar magnitude, but costs are likely to vary considerably depending on building type and level of protection desired.

Retrofit of existing buildings is considerably more expensive. The retrofit cost in the Istanbul base case above is assumed to be 32 percent of the rebuilding costs (see figures in Wesiner et. al., 2004). Two different approaches to apartment building retrofit retrofit in Istanbul are priced at 19% (external) and 39% (internal) by Sucuoglu et al. (2006). Some retrofit estimates climb above 50 percent (Smyth et. al. 2004b). The cost of retrofitting 3,600 public structures in Istanbul was estimated at $1 billion – or approximately $280,000 per structure (World Bank, 2005). Retrofitting is also technically complex, and can be botched as easily as the construction of a new building designed to withstand earthquakes in the first place (Jain, 2005). This suggests that the 'true cost' per measure of earthquake safety actually achieved (rather than earthquake safety paid for), may be even higher than the estimated costs.[11]

Regarding the probability of a destructive natural disaster hitting a particular area, even risk-prone regions very rarely face a major natural disaster because the major impact of natural disasters each year is concentrated in a tiny proportion of the territory at risk from such events (see Box 6). Given that, and the comparatively limited and recent nature of

[11] The problem of low returns to investment is a significant one throughout the developing world. Indeed, a considerable macroeconomic literature points out that they only way one can explain Africa's low growth despite comparatively high investment over the last fifty years is by the fact that much of the investment resources were squandered on building the wrong thing and building it badly (Pritchett, 1996).

(Smyth et al., 2004b). The base case above uses a $250,000 value. One method to come up with the 'value of a statistical life' for benefit-cost is contingent valuation which attempts to place a value on life by asking people or estimating indirectly how much they would pay to avoid situations that might lead to injury or death. Using a range of contingent valuation methods, the value of a statistical life has been estimated at $519,000 to 675,000 in Santiago Chile, $413,000 in Taiwan and $250,000 in rural Thailand, for example. As a rule, values are lower in poorer countries – people have fewer resources with which to avoid risky situations (Viscusi and Aldy, 2003, Liu et, al., 1997, Bowland and Beghin, 2001, Gibson et. al., 2007). One recent analysis suggests that a rule of thumb value of a statistical life would be 60 times incomes in low-income countries (Cropper and Sahin, 2008). This suggests numbers as low as $30,000 in the very poorest countries. All of these methods are open to both methodological and moral critique, but they do suggest that, for most developing countries at least, $1 million is an unjustifiably high figure, even if it appears low relative to safety at work requirements in the US.

A different and perhaps less contentious way to look at the issue of the efficacy of the health benefits of retrofit is to think about the relative cost effectiveness of other interventions which save lives. Using the base case above, over thirty years, the average retrofit saves a life at a cost of $135,000. This is a considerably lower estimate than the $430,000 figure that we saw for commercial buildings in the US. Regardless, this number suggests an approximate cost per disability-adjusted life year of $2,600.[18] Note that this does not include the benefit of injuries avoided by building collapse –but even if avoided disability accounts for a considerable part of total disability-adjusted life years saved, costs are still considerably higher than a range of other interventions. In developing countries, millions of people die each year from diseases that can be cured using a simple regime of oral antibiotics, which costs as little as $0.25. More broadly, there are a range of interventions that cost less than $2 per disability-adjusted life year saved (Laxminarayan et. al. 2006, Boone and Zhan, 2006). These interventions are far from universally applied in developing countries and in terms of cost efficiency they carry significantly higher health benefits per dollar than retrofit may. In terms of priorities, then, resources dedicated to improving health outcomes should probably be applied first to other interventions than building retrofit.

The focus of the above discussion has been on lives lost. In both human and economic terms, this is probably appropriate for earthquakes in developing countries, where loss of life is the major impact –and especially appropriate when discussing apartment buildings. However, such an approach will miss many of the costs of disasters and the benefits of mitigation measures. In the case of other natural disasters, floods may kill fewer people, but they destroy considerable property. In richer countries, the marginal mitigation measure is already designed to save property, because existing regulations (largely) protect against loss of life. The Northridge earthquake in California killed 72 people, but cost $6.5 billion in terms of business interruption losses, for example (Mechler, 2005). In the case of commercial or government buildings, there is also a considerable cost of

[18] Assuming the average earthquake victim would otherwise enjoyed fifty years of good health, and not allowing for discounting.

business interruption. And even with apartment buildings, there is the cost of personal property lost and the need to house survivors during reconstruction (although building retrofit itself is also considerably disruptive and may require moving occupants and this is a certain cost we have not accounted for).

Once we have accounted for costs and benefits, given that large-scale disasters are rare, the outcomes of benefit-cost calculation regarding risk reduction for such disasters depend crucially on the discount rate adopted. This is not the place for an extended discussion of the appropriate discount rate to use,[19] but Box 7 suggests that it may be suitable to use higher discount rates for health outcomes in developing countries than it is in the developed world.

Box 7: What discount rate for disaster risk reduction?

Low discount rates may be appropriate if we believe that we should value people in 100 years as much as we value ourselves –this may be a good moral position but not one that individuals tend to follow. Even for public mitigation projects, If we believe that we can generate an economic rate of return of ten percent on some other project, it would be worth investing in that project and using the proceeds to fund a cost-effective health intervention rather than investing in mitigation. In other words, the benefit-cost ratio of mitigation may be positive using a low discount rate, but a low discount rate may also increase the benefit-cost ratio of a number of other investments, which should (still) be undertaken first.

Furthermore, for developing countries it is worth noting that a higher discount rate to risk mitigation measures is rational. Oster (2005, 2006) argues that responses to the AIDS epidemic in Sub-Saharan Africa can be better understood if we take account of the broader risk profile faced by potential victims. Health risks of all types are considerably higher in developing countries, and so the marginal impact of an additional health risk is lower (‘if AIDS doesn’t get you, malaria will’). A similar rational resignation may play a role in decisions regarding mitigation measures which may only pay off over a thirty or forty year time horizon.

Given that all of the assumptions required to make a cost-benefit analysis of the benefits of retrofit (or other disaster risk reduction investments) are open to considerable uncertainty and debate, it is perhaps safe to say that a strong conclusion in favor of an intervention would need to be based on a large benefit-cost ratio robust to conservative assumptions regarding benefits. A number of existing benefit-cost analyses in the literature regarding retrofitting may not have passed this test. Box 8 discusses two additional exercises in calculating the benefit-cost of retrofit in the case of housing.

The financial proposition facing the individual home or business owner or the government regulator or minister is extremely complex, then – involving high up-front costs for benefits likely to accrue over the long term if they accrue at all. In the aftermath

[19] See Stern (2006) along with Richard Tol’s response: http://www.fcpp.org/pdf/CritiqueofSternReport.pdf or William Nordhaus’ comments: http://nordhaus.econ.yale.edu/SternReviewD2.pdf for a discussion of the discount rate as regards climate change risk reduction, and a discussion of the ethical implications of Stern’s discounting methodology in Kenny (2007).

how robustly positive such ratios are will depend on the same range of assumptions and estimates that we encountered above.

And despite undoubted benefits to resistant school construction, for example, it is worth comparing the cost of resistant construction to school budgets in developing countries. If school retrofit/earthquake proof construction comes at the cost of building fewer schools and excluding girls and boys from education, the long-term health effects of such construction may even be negative (See Box 9). It is worth noting that even while as many as 2,500 children worldwide die each year in school collapse, more than 10,000,000 children under the age of five die each year from other causes before they can even make it to school – and the majority of those deaths can be easily and cheaply prevented. In areas of very high risk of earthquake, where cheap engineering solutions are available, the benefit to cost ratio of such projects can look very good. If the risk of earthquake is lower, or costs higher, retrofit in particular may look less attractive than other methods of improving child health. This is particularly the case where overall levels of child health are poor, and particularly in cases where earthquake proofing will come at the expense of additional school construction and the exclusion of children from educational opportunities.

A more detailed analysis based on school retrofit in Istanbul suggested that the program would make sense at a value of $400,000 per life saved. [23] Compare this number to education expenditures in poor countries. 'Discretionary' expenditure – money left over after paying teachers' salaries used to cover supplies, teaching equipment, utility bills, building maintenance and construction—is as low as $5 per year per primary student per year in many countries (Grace and Kenny, 2003). If the average school size is 100 children, this suggests retrofitting alone would be equal to 16 years of a school's total discretionary expenditure in many poor countries. This expenditure will actually benefit only some small proportion of the schools which embark on it (in preventing collapse that otherwise would have occurred).

Perhaps more to the point, it is worth comparing the costs of improved school building safety to a number of other efforts to improve health outcomes in developing countries – which we have seen may save lives at far lower costs. When we add in the relative simplicity of the health interventions compared to the complex engineering often required for retrofit, likely benefits appear to skew ever more heavily to health expenditures over retrofitting. As noted in some cost benefit studies of the value of retrofitting projects, it is not uncommon to find that existing construction is substandard and quality control of building materials was limited (Smyth et. al. 2004a). These same problems will reduce the effectiveness of retrofitting projects, thereby (and perhaps severely) reducing benefit-cost ratios.

[23] This exercise by Smyth et. al. (2004a) suggests retrofit costs between $8-20,000, the loss of fifteen lives in the event of collapse, a replacement cost of $160,000 and a value of a statistical life of $400,000. Using a 3% discount rate, assuming that schools are occupied one third of the time (apparently not allowing for holidays and weekends) and earthquake risk calculated for Istanbul, the estimate produces strongly positive net present values. For reasons discussed in the text, some of these estimates and assumptions may be more favorable than justified. Nonetheless, this analysis does suggest that in higher-risk areas such as Istanbul, school retrofitting projects may be justifiable.

This is not to say that all earthquake mitigation measures regarding public building construction and retrofit in developing countries are an inefficient use of resources. It does suggest the importance of a focus on the most vulnerable schools and buildings, alongside those buildings that are most important in the aftermath of a disaster (not least hospitals) as well as recognition of the tradeoffs involved in expenditure on retrofitting or robust construction.[25] It also suggests the importance of an initial focus on disaster risk reduction around the most efficient measures which may involve disaster planning, emergency communications and public infrastructure measures rather than retrofit of individual buildings.

Indeed, to focus on retrofit or even initial design against earthquake damage in individual buildings is to look at some of the disaster risk reduction measures with the lowest benefit-cost ratios, however. Higher benefit-cost ratios perhaps unsurprisingly appear to accrue to risk reduction measures against comparatively common but relatively small scale disaster events – local flooding, for example. We have seen that the large-scale disasters that cause the most damage and kill the most people are (thankfully) particularly rare, and this very rarity reduces the benefit-cost ratios associated with risk reduction.

For FEMA mitigation projects in the US the benefit-cost ratio of flood mitigation projects is estimated at an average of around 5 compared to 1.5 for the average earthquake mitigation measure (OECD, 2008). The available evidence suggests higher benefit-cost ratios to measures that involve preparedness, community-level activities or stand-alone engineering solutions (flood walls, for example) rather than changed construction practices for all buildings or retrofit.

Such government-sponsored community-level disaster risk reduction projects in a range of settings have demonstrated high benefit-cost ratios.[26] In particular, there is considerable evidence of quite high returns to programs that emphasize planning and community-level preparedness for floods and cyclones – two studies of such projects in India, which involved construction of an escape route, provision of boats for evacuation, raised water pumps, a village rescue and evacuation team and a flood evacuation plan had benefit-cost ratios estimated between 3 and 20 (ERM, 2005). Similarly, the Bangladesh Cyclone Preparedness Programme (CPP) includes shelters, early warning systems and community-based preparedness measures. The annual operating cost is $460,000. While it is hard to fully evaluate the effectiveness of the program, it is worth noting the loss of life in three cyclones of similar strength that hit Bangladesh in 1970 (prior to the CPP) and 1991. The 1970 cyclone affected 3.6 million people and killed 300,000 people. The 1991 cyclone affected far more – 15 million people. But it killed 138,000 – around a third of the number killed in 1970. A similar strength cyclone in 1997 affected 3 million people – nearly the same number as 1970. But only 11 people died (ERM, 2005).

[25] The October 2005 earthquake in Pakistan destroyed 50 percent of health facilities in affected areas, surely increasing the death toll from the disaster (Bhatia, 2008b).
[26] In the US, one estimate suggests that every dollar spent by FEMA on mitigation grants achieves savings in terms of avoided grants and tax revenue losses of $3.65 (Burby, 2008).

Public engineering efforts to reduce disaster risk have also demonstrated cost-effectiveness. A World Bank appraisal of a flood protection project in Argentina suggested an internal rate of return of 20 percent, and a range of other flood and hurricane protection projects in the developing world (in countries including the Philippines, Argentina, Peru, Indonesia, China and Brazil) have estimated cost-benefit ratios of two and above (Moench et. al. 2007, see also OECD, 2008, Healy, 2008, Mechler, 2005).[27]

The Role of Irrationalities, Market and Government Failures

Retrofit of private homes against earthquakes may have some of the lowest returns in the arsenal of disaster risk reduction measures, then – but even these measures may be priority investments in comparatively wealthy communities at high risk from large earthquakes and low existing building standards. And other mitigation measures involving public construction protecting against more common disaster risks, for example, carry higher benefit-cost ratios. Yet many such measures remain unimplemented.

It is unlikely that rational, fully informed calculations of the costs and benefits of the whole range of mitigation measures are the sole (or perhaps even the primary) reason for their limited adoption by individuals or governments, then. Greater expenditure on mitigation measures will make more sense in areas where natural disasters are particularly common, and yet the depressing result of cross-country analysis is that, allowing for income and other factors, areas repeatedly exposed to disasters do not see lower per-disaster mortality rates, as we have seen. This suggests mitigation measures are underutilized even where they would carry the highest returns.

It may be that the rational agent model fails with rare catastrophic events, and that consumers and citizens do not correctly evaluate the costs and benefits of (in)action regarding 'acts of God' (Praeter and Lindell, 2000). A number of common irrational decision-making devices used by individuals can impact the level of risk-reduction effort undertaken. Experimental and natural observation suggest that people use budgeting heuristics which pressure against mitigation spending, that they display biases in inter-temporal planning and chose inaction in the face of uncertainty, and that they ignore the risks of an event (however catastrophic it might be) if the event is seen as below some threshold of probability (Camerer and Kunreuther, 1989). It is very difficult to calculate the benefits accorded by mitigation as we have seen, suggesting an information gap. Combined with the human tendency to remain passive in the face of uncertain events, this may encourage inadequate mitigation response. For individuals and businesses, there may be excessive faith in government mitigation measures, which further reduces their own incentive to act (Box 10).

[27] Although studies do suggest that further flood control measures in Japan would carry very low benefit-cost ratios (OECD, 2008).

disasters. In this regard, it is worth noting that, concerns regarding benefit-cost aside, many countries do have building codes on the books that set standards for earthquake-resistant construction – and yet buildings routinely collapse in earthquakes. This suggests an important additional element to the story regarding the political economy of the implementation of regulation.

The Political Economy of Regulation

That consumers are ill-equipped to verify the quality of construction may account for the pressure for government regulation. The information asymmetry could be overcome by private agencies – on the model of credit rating agencies, for example. In public buildings in particular such a model would be likely unworkable as a tool to improve safety. Imagine an office worker relocated to an office with a higher 'earthquake risk rating' because of lower quality construction. Would it be plausible for them to quit? Or imagine shoppers entering a store pausing to check the earthquake risk rating register to evaluate the chance they would die while in the shop. For private houses, the risk rating model might be more plausible – indeed, publicly provided information on location-based earthquake risk is part of the approach used in Tokyo and California, as we have seen. That there is limited consumer response in these cases may be one reason why public safety advocates (and risk-averse consumers worried they won't get a return from their safety investments) push for regulation. In addition, insurance and construction firms in strong governance environments may both benefit from such regulation. Insurers can be guaranteed a certain quality of construction in the buildings they insure without the cost of assessment. Contractors firms gain additional business from retrofit and earthquake-proof construction.[29]

The incentives for insurers and contractors to push for such regulation are lower in weaker-governance economies. Owners can avoid regulation because of weak enforcement or bribery. So insurers can't be confident that buildings will be constructed according to code. And contractors may lose out to informal construction techniques or less scrupulous firms if they insist on building to code. But for a corrupt senior official in charge of construction, passage and non-enforcement of codes may be an optimal approach to the regulation of construction. Overseeing an enforcement regime allows for the collection of payments from construction firms that flout the codes. The complexity of construction will mean that shoddy construction is unlikely to be detected in the short term. Buildings may subsequently fail but this is likely to be some time in the future when the official has moved on to another position. Regardless, the official can blame limited capacity or construction firm obfuscation and is unlikely to face significant risk of penalty.

These factors may explain why poorer countries are less likely to have regulations which comply with international earthquake design codes and standards, and less likely to enforce them if they are on the books (Anbarci, Escaleras and Register, 2005). A review of natural hazard-related building codes in developing countries found that the existence

[29] Thanks to Phil Keefer for this observation.

of such codes varied significantly across countries –the two countries surveyed in Africa had a total lack of hazard-related codes (World Bank 2008).[30] Even where codes existed, they were frequently hard to access and their legal status was often unclear (as to being a guideline or a legal requirement).

Istanbul lacked significant seismic building codes until 1997, but even when enacted, Istanbul's codes were rarely enforced – which suggests an additional problem with regulatory institutions. The Marmara quake in Turkey which killed 17,000 was of a magnitude and type accounted for by existing design specifications in the Turkish seismic code (Sezen et. al. 2002) – it was not lack of regulation but lack of enforcement which led to deaths. In Istanbul prior to the quake 'amnesties' for buildings known to be sub-code were passed repeatedly, and municipalities responsible for enforcement lacked the capacity to do so. An estimated 70 percent of housing did not conform with extant standards of resistance to seismic risk (Escaleras, Anbarci and Register, 2007, Anbarci, Escaleras and Register, 2005, World Bank, 2005).

Problems went beyond official amnesties to limited capacities and rent-seeking. Correct construction of reinforced buildings is technically complex, and in many developing countries the requisite skills are scarce not only to construct but also to monitor construction (Anbarci, Escaleras and Register, 2005).[31] Municipalities including Istanbul had weak and underfunded municipal engineering and planning departments staffed with unaccredited engineers prone to corruption. In 2006, 40 municipal officials in three towns in Turkey were arrested for taking bribes in return for allowing unlicensed construction (Escaleras, Anbarci and Register, 2007)

Again, this problem is far from unique to Turkey. Cross-country statistical analysis by Escaleras, Anbarci and Register (2007) suggests that, allowing for income and a range of other factors, countries perceived as more corrupt see higher earthquake fatalities. Jain (2005) discussing "the Indian earthquake problem" suggests why this may be the case. He notes that seismic code building certificates are "easy to procure, sometimes on payment of small money, and need not have any correlation with how a building is built."[32] After the 2001 Bhuj quake in India, it emerged that no on-site inspection of the

[30] This is not to say the problem is unique to developing countries. Counties and cities in the US located in states which have adopted comprehensive land planning requirements for local government have seen lower national flood insurance program claims and payments. This is true even if the requirements do not explicitly require attention to natural hazards. Despite this, less than half of US states have adopted such planning requirements (Burby, 2006).

[31] Again, this is not a problem specific to developing countries. Dade County, Florida which suffered the greatest damage during Hurricane Andrew in 1992, had 60 building inspectors to police 20,000 new buildings constructed each year –suggesting an average of 35 inspections per inspector per day (IRC and IIPLR, 1995). Furthermore, as many as one half of local building officials working on the Gulf coast did not understand or did not enforce hurricane-related codes in 1992. Dade County's failure to enforce codes alone accounted for $4 billion of additional damage to insured buildings caused by Hurricane Andrew. (Burby, 2006)

[32] It is worth noting that, at least within countries, the causal relationship between corruption and natural disasters may run both ways. Leeson and Sobel (2008) find that in the United States, states which have more natural disasters see more corruption-related convictions per capita. They provide some evidence that this is because poorly managed windfalls in the form of Federal Emergency Management Agency relief

Box 11: Does regulation improve construction outcomes in developing countries?

Weak oversight and corruption in construction regulation may help to account for the apparently fragile relationship between regulations on the books and outcomes on the ground. *Doing Business* surveys suggest that the number of procedures required in order to get permission to build a warehouse varies considerably between countries. The average number of procedures was 16 in rich countries, and 20 in low income countries, with the time taken to comply 157 days in rich countries and 229 in poor countries. It is not clear that this extra regulatory burden is improving outcomes -there is no correlation between the number of procedures and the number of worker accidents across countries, for example. More broadly, greater regulation in countries at similar levels of income is associated with lower compliance with international quality standards as measured by ISO 9000 compliance, more perceived corruption and a larger informal economy (Djankov, La Porta Lopez-de-Silanes and Shleifer, 2000).

These results should not be taken to suggest that regulation is an unnecessary burden in developing countries, but it should instill caution regarding solutions which involve even greater regulation –will that regulation be enforced fairly, or will it act as one more source of rents for officials, with little impact on quality and safety?

Given the high cost of complying with regulations regarding retrofit and earthquake proofing, their comparatively low benefit, and the significant negative side-effects of such regulation in environments prone to corruption, the optimal solution may be to focus, simplify and limit such regulation – alongside similarly complex rules regarding land use. In particular it might be best to focus limited regulation and oversight capacity in areas where potential returns are highest – large public buildings, for example.

The Political Economy of Public Construction Projects

In addition to their role as regulators, government officials are the largest customers for construction firms around the world. Given the repeated accusations regarding the particular fragility of government buildings such as schools and hospitals in recent earthquakes, it appears likely that there are a particularly acute set of political economy factors related to such construction.

Benefit-cost analysis may bring further clarity to the tradeoffs presented to an official deciding on construction methods as much as for private individuals. As it is more expensive to construct earthquake-resistant buildings, the official faces a choice between construction of more buildings with a risk of collapse at some point in the next 100 years or construction of fewer buildings with lower risk of collapse. These costs of mitigation are immediate, then, but the visibility of such measures is low, and any benefits are likely to accrue to a politician's successors. And we have seen that public appetite for such construction appears to be limited.[37]

[37] An additional element that applies to local construction projects is the potential moral hazard presented by levels of government. Local governments may feel able to skimp on mitigation on the implicit